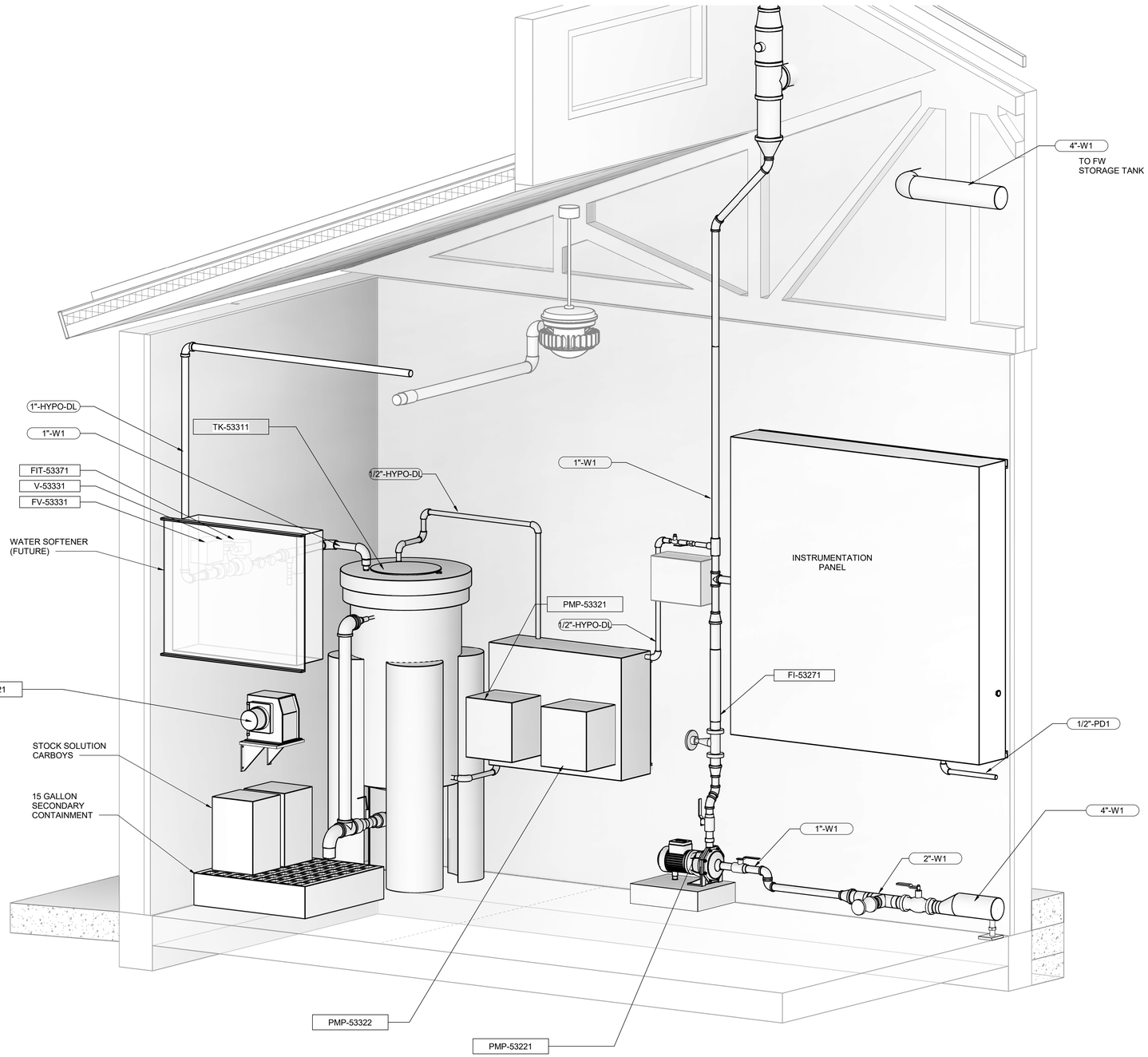


ISOMETRIC VIEW-CHEMICAL AREA

SCALE (A)

SCALE (A) 1 0 1 2 3 3/4" = 1'-0"

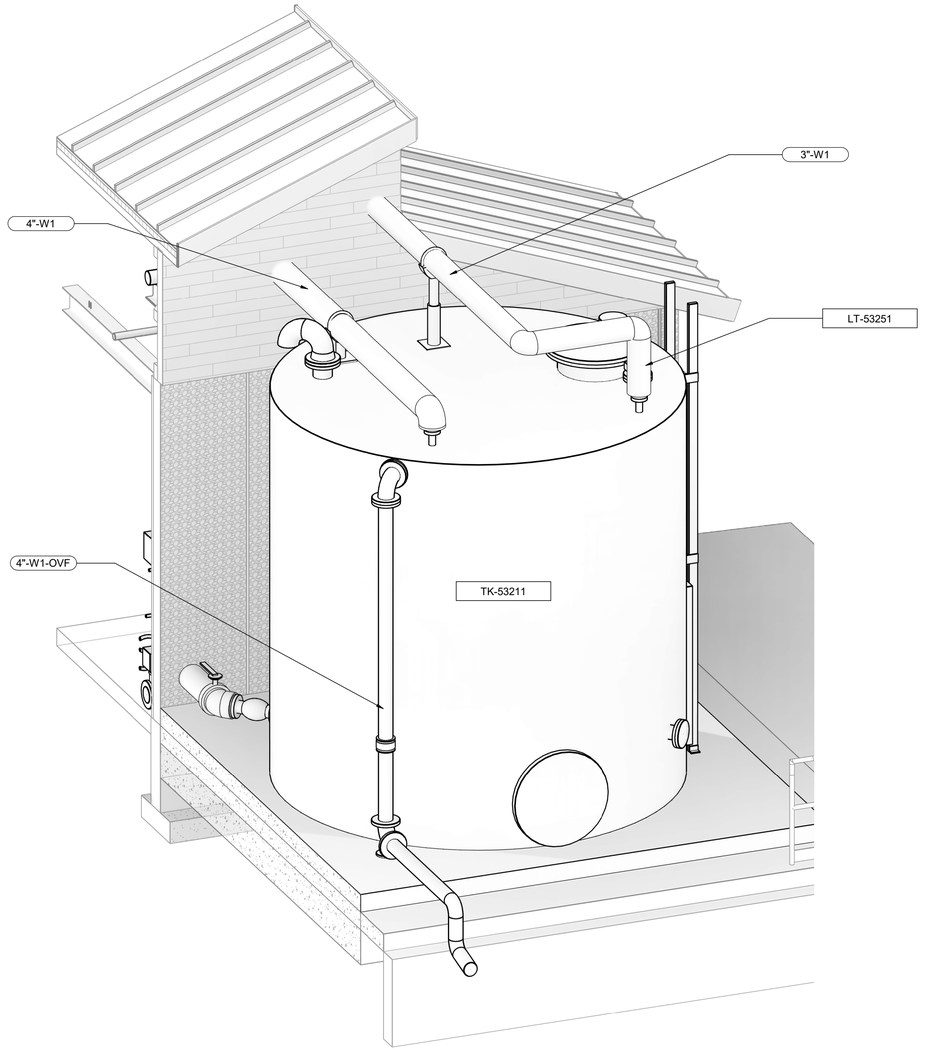
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D904	TITLE OF SHEET RSPH CHEMICAL AREA ISOMETRIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 79 OF 113

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ISOMETRIC VIEW-FINISHED WATER STORAGE TANK

SCALE (A)

GENERAL NOTES

A. SEE SHEET 005D505 FOR PIPE PENETRATION DETAILS.

SCALE (A) 2 0 2 4 1/2" = 1'-0"

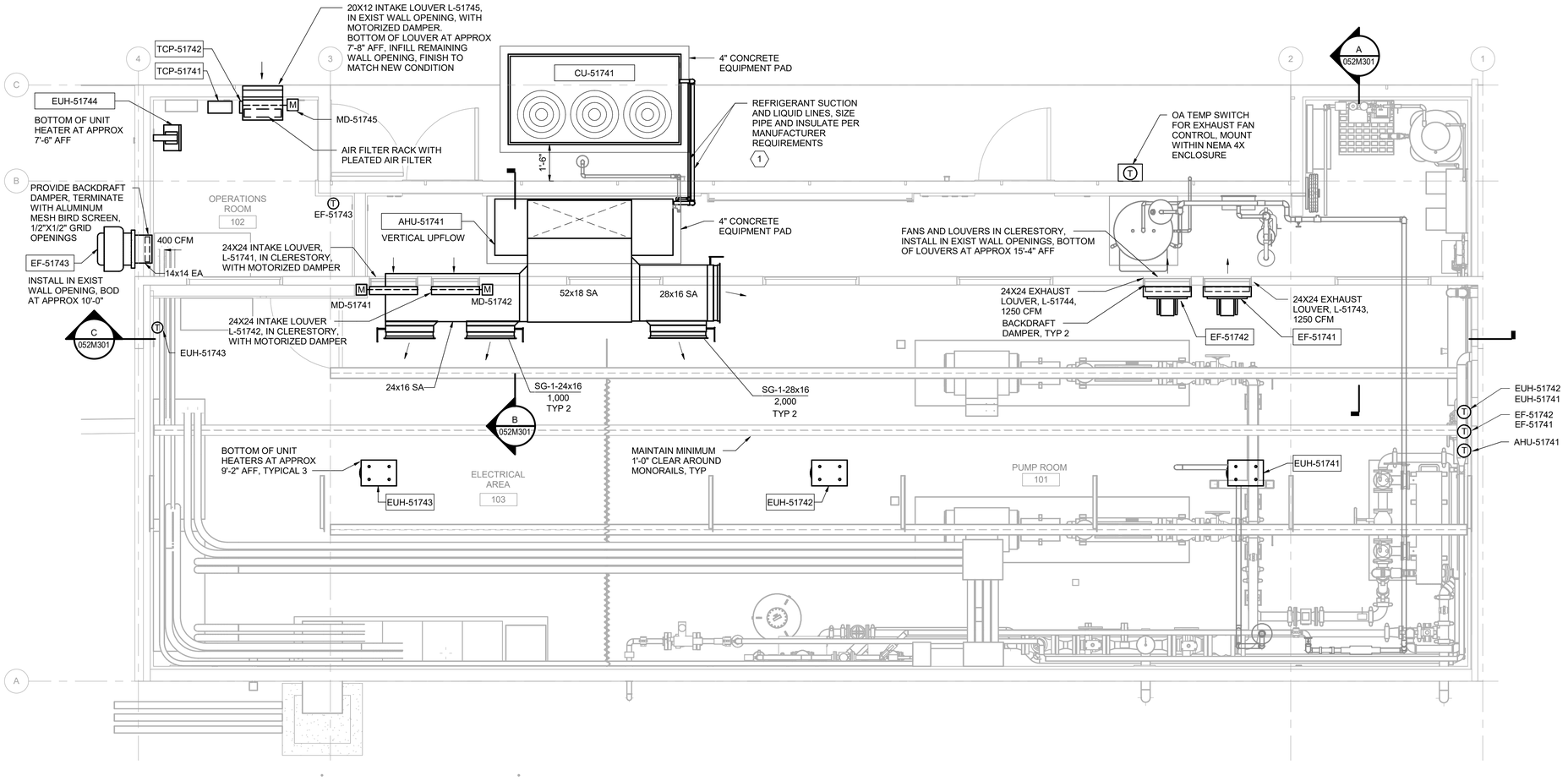
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAFTED: C. CONFER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 D905	TITLE OF SHEET RSPH FINISHED WATER STORAGE TANK ISOMETRIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 80 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_D.rvt
4/27/2026 5:23:13 PM



KEYNOTES: (X)

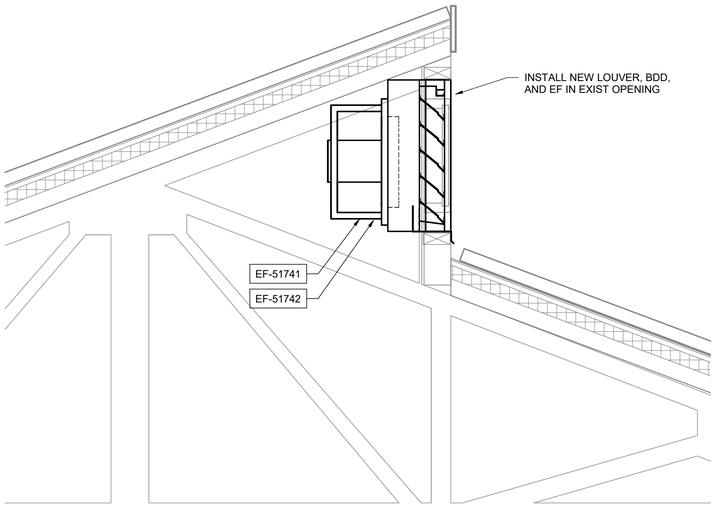
1. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

1 / 052M101 HVAC PLAN
SCALE (A)

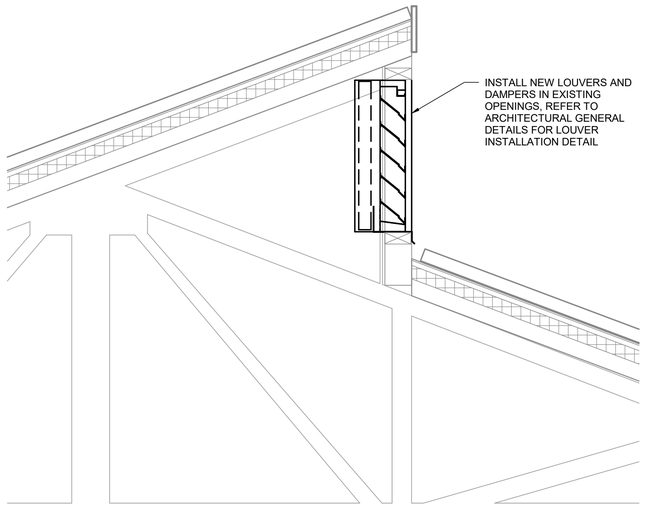
SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 052 M101	TITLE OF SHEET RSPH HVAC PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 81 OF 113

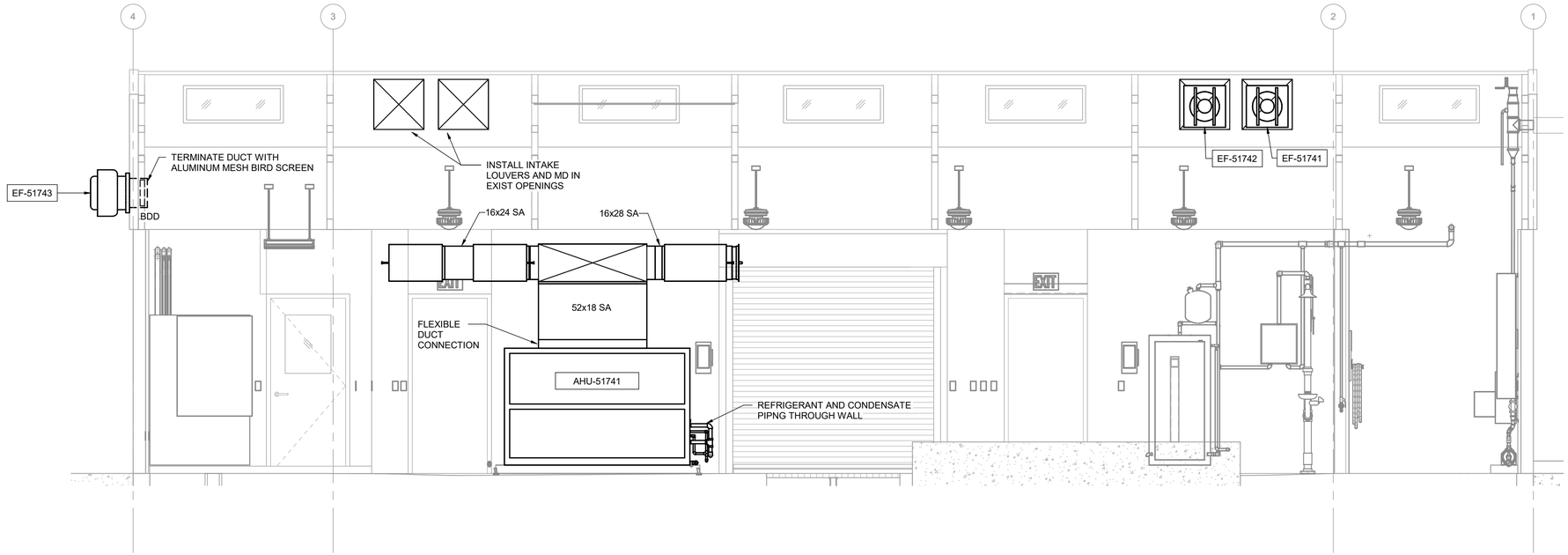
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A 052M101 PUMP ROOM CLERESTORY - EXHAUST FANS
SCALE B



B 052M101 PUMP ROOM CLERESTORY - LOUVERS
SCALE B



C 052M101 PUMP ROOM
SCALE A

SCALE B 1 0 1 2 1" = 1'-0"
SCALE OF FEET

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_MP.rvt
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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 052 M301	TITLE OF SHEET RSPH HVAC SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 82 OF 113

AIR HANDLING UNIT SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	TYPE	ASSOCIATED OUTDOOR UNIT	COOLING COIL			SUPPLY FAN					ELECTRICAL			OPERATING WEIGHT (LBS)	NOTES
						TOTAL COOLING CAPACITY (MBH)	SENSIBLE COOLING CAPACITY (MBH)	LAT (DB, °F)	AIRFLOW (CFM)	EXTERNAL SP (IN. W.C.)	FAN RPM	MOTOR BHP	MOTOR HP	V/HZ/PH	MCA	MOCP		
AHU-51741	CARRIER 40RLAA16	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	COOLING ONLY, VERTICAL UPLFLOW	CU-51741	152.0	152.0	57.0	6,000	0.60	840	2.40	3.7	480/60/3	12.0	20	713	1,2,3,4,5

NOTES:

1) UNIT PERFORMANCE IS LISTED AT SITE SPECIFIC DESIGN TEMPERATURES AND ELEVATION.
2) PROVIDE SUB-BASE MOUNTING FRAME AND ANCHOR TO CONCRETE PAD.
3) PROVIDE AIR FILTER RACK WITH 2" MERV-8 PLEATED FILTERS.
4) PROVIDE A 2-SPEED FAN VFD FOR TWO STAGES OF COOLING.
5) PROVIDE NEOPRENE VIBRATION ISOLATORS AT BASE OF UNIT.

GENERAL NOTES:

A. MECHANICAL SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF MECHANICAL EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

CONDENSING UNIT SCHEDULE (COOLING ONLY)

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	INDOOR UNIT SERVED	TOTAL COOLING CAPACITY (BTUH)	REFRIGERANT	EER	ELECTRICAL			OPERATING WEIGHT (LBS)	NOTES
								V/HZ/PH	MCA	MOCP		
CU-51741	CARRIER 38AXDT16	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	AHU-51741	152,000	R-454B	11.0	480/60/3	29	40	731	1,2,3,4

NOTES:

1) COOLING CAPACITY IS LISTED AT SITE SPECIFIC DESIGN TEMPERATURES AND ELEVATION.
2) PROVIDE E-COAT EPOXY COATING ON CONDENSER COIL FOR CORROSION PROTECTION.
3) PROVIDE HAIL GUARDS TO PROTECT THE COIL.
4) PROVIDE MINIMUM 2 STAGES OF COOLING.

SITE CONDITIONS

SITE LOCATION	GRAND CANYON NATIONAL PARK, NORTH RIM
	COCONINO COUNTY
ELEVATION ABOVE SEA LEVEL (FT)	ROARING SPRINGS PUMP HOUSE: 4870'
	WTF / WWTF: 8,300 FT
ASHRAE CLIMATE ZONE	5B (PER 2021 IECC)
OUTDOOR DESIGN TEMPERATURES	SUMMER (DB/WB): 90 / 56 °F
	WINTER (DB): -1 °F

NOTES:

1) WEATHER DATA IS SOURCED FROM THE 2021 ASHRAE FUNDAMENTALS ONLINE CLIMATE DATABASE FOR 0.4% (SUMMER) AND 99.6% (WINTER) CONDITIONS FOR GRAND CANYON NATIONAL PARK, ARIZONA.

FAN SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	DESCRIPTION	AIRFLOW (CFM)	EXTERNAL SP (IN. WG.)	DRIVE TYPE	FAN RPM	ELECTRICAL			OPERATING WEIGHT (LBS)	NOTES
									V/HZ/PH	MOTOR HP	BHP		
EF-51741	GREENHECK SE1-14-432-VG	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	WALL MOUNTED PANEL FAN	1,250	0.2	DIRECT	1,279	115/60/1	1/4	0.09	24	1,2
EF-51742	GREENHECK SE1-14-432-VG	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	WALL MOUNTED PANEL FAN	1,250	0.2	DIRECT	1,279	115/60/1	1/4	0.09	24	1,2
EF-51743	GREENHECK CUE-090-E	ROARING SPRINGS PUMPHOUSE	OPERATIONS ROOM	WALL MOUNTED EXHAUST FAN	400	0.125	DIRECT	1,213	115/60/1	1/10	0.03	47	1,2,3

NOTES:

1) PROVIDE WITH WALL MOUNTING COLLAR, OSHA FAN/MOTOR GUARD AND BACKDRAFT DAMPER.
2) PROVIDE ELECTRICALLY-COMMUTATED MOTOR WITH MOTOR-MOUNTED SPEED CONTROL DIAL.
3) PROVIDE WITH WALL-MOUNTING BRACKET AND BACKDRAFT DAMPER.

DESIGN TEMPERATURES

AREA	SUMMER DB (°F)	WINTER DB (°F)
OPERATIONS ROOM	UNCONDITIONED	55
PUMP ROOM	80	55

LOUVER SCHEDULE

TAG NO.	BUILDING SERVED	AREA SERVED	FUNCTION	AIRFLOW (CFM)	LOUVER DIMENSIONS (IN)			NOTES
					WIDTH	HEIGHT	DEPTH	
L-51741	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	INTAKE	1,250	24	24	4	1,2,3
L-51742	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	INTAKE	1,250	24	24	4	1,2,3
L-51743	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	EXHAUST	1,250	24	24	4	2,3
L-51744	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	EXHAUST	1,250	24	24	4	2,3
L-51745	ROARING SPRINGS PUMPHOUSE	OPERATIONS ROOM	INTAKE	400	20	12	4	1,2,3

NOTES:

1) PROVIDE WITH INTEGRAL MOTORIZED DAMPER AND 115V DAMPER ACTUATOR.
2) MINIMUM 50% FREE AREA.
3) MAX PRESSURE DROP OF 0.1 INCHES WATER COLUMN AT DESIGN AIRFLOW.

DIFFUSERS, GRILLES, REGISTERS

TAG NO.	BASIS OF DESIGN MAKE/MODEL	AREA SERVED	DESCRIPTION	FUCTION	MOUNT	MATERIAL	FINISH	NOTES
SG-1	TITUS 272FS	GENERAL	RECTANGULAR	SUPPLY	DUCT	ALUMINUM	WHITE	1,2,3,4

NOTES:

1) PROVIDE TWO SETS OF BLADES FOR DOUBLE-DEFLECTION OF AIRFLOW.
2) MAXIMUM TOTAL PRESSURE DROP OF 0.12 INCHES WATER COLUMN ACROSS GRILLE OR DIFFUSER AT DESIGN AIRFLOW.
3) PROVIDE WITH INTEGRAL OPPOSED BLADE DAMPER, OR SEPARATE DUCT-MOUNTED DAMPER FOR BALANCING.
4) ADJUST BLADES DURING FINAL TEST AND BALANCE TO PROVIDE EVEN DISTRIBUTION OF AIRFLOW IN THE SPACES SERVED.

ELECTRIC UNIT HEATER SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	AIRFLOW (CFM)	HEATING CAPACITY (KW)	ELECTRICAL		WEIGHT (LBS)	NOTES
						V/HZ/PH	TOTAL AMPS		
EUH-51741	INDEECO UHIR 240-U1037U	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	350	3.7	480/60/3	4.66	25	1,2,4,6
EUH-51742	INDEECO UHIR 240-U1037U	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	350	3.7	480/60/3	4.66	25	1,2,4,6
EUH-51743	INDEECO UHIR 240-U1037U	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	350	3.7	480/60/3	4.66	25	1,2,4,6
EUH-51744	INDEECO UHIR 240-U1025C	ROARING SPRINGS PUMP HOUSE	OPERATIONS ROOM	350	2.5	208/60/1	12.3	25	3,5,6

NOTES:

1) FURNISH WITH CEILING MOUNTING KIT.
2) FURNISH WITH MANUFACTURER ANEMOSTAT DIFFUSER.
3) FURNISH WITH WALL MOUNTING KIT.
4) PROVIDE WITH WALL-MOUNTED THERMOSTAT. PROVIDE WITH CONTROL TRANSFORMER AND LOW VOLTAGE CONTROL CIRCUIT.
5) PROVIDE WITH UNIT-MOUNTED THERMOSTAT.
6) PROVIDE WITH MANUFACTURERS DISCONNECT SWITCH.

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

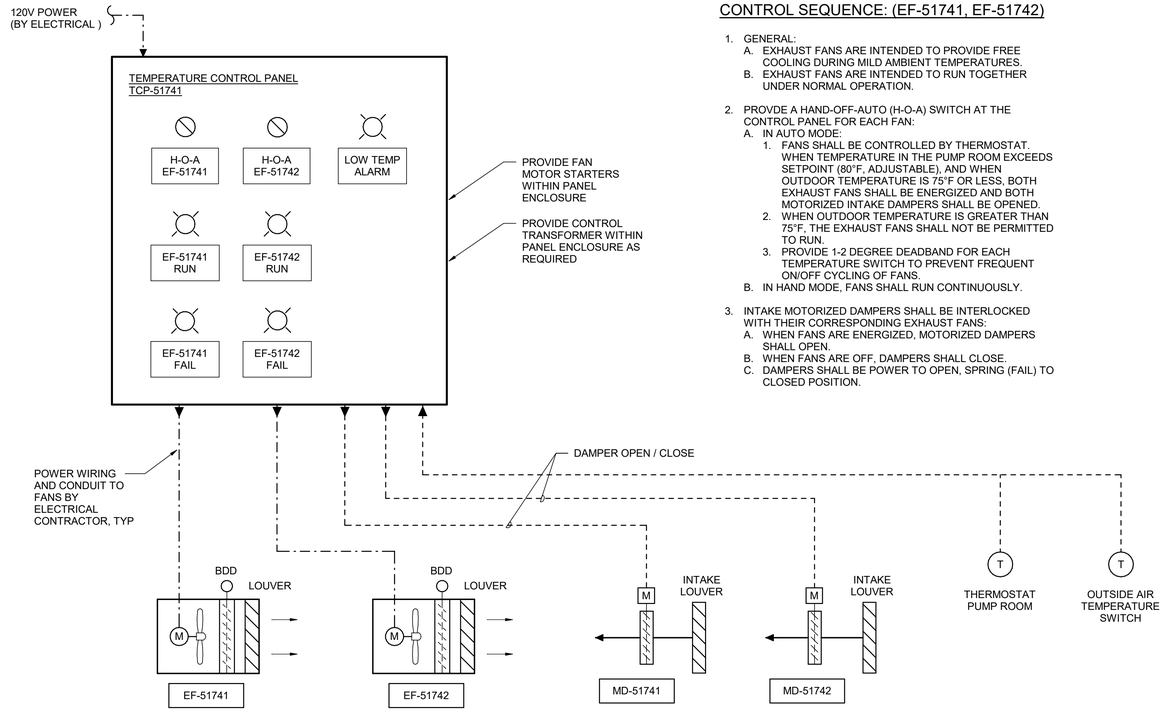
DESIGNED: A. PUTMAN
CADD: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 052 M601

TITLE OF SHEET
RSPH
HVAC SCHEDULES

NORTH RIM - GRAND CANYON NATIONAL PARK

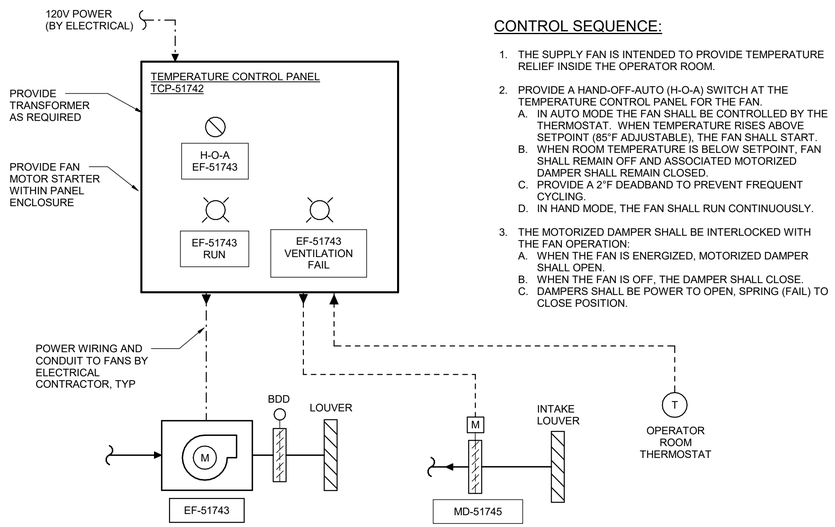
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 83 OF 113



CONTROL SEQUENCE: (EF-51741, EF-52742)

1. GENERAL:
   A. EXHAUST FANS ARE INTENDED TO PROVIDE FREE COOLING DURING MILD AMBIENT TEMPERATURES.
   B. EXHAUST FANS ARE INTENDED TO RUN TOGETHER UNDER NORMAL OPERATION.
2. PROVDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE CONTROL PANEL FOR EACH FAN:
   A. IN AUTO MODE:
      1. FANS SHALL BE CONTROLLED BY THERMOSTAT. WHEN TEMPERATURE IN THE PUMP ROOM EXCEEDS SETPOINT (80°F, ADJUSTABLE), AND WHEN OUTDOOR TEMPERATURE IS 75°F OR LESS, BOTH EXHAUST FANS SHALL BE ENERGIZED AND BOTH MOTORIZED INTAKE DAMPERS SHALL BE OPENED.
      2. WHEN OUTDOOR TEMPERATURE IS GREATER THAN 75°F, THE EXHAUST FANS SHALL NOT BE PERMITTED TO RUN.
      3. PROVIDE 1-2 DEGREE DEADBAND FOR EACH TEMPERATURE SWITCH TO PREVENT FREQUENT ON/OFF CYCLING OF FANS.
   B. IN HAND MODE, FANS SHALL RUN CONTINUOUSLY.
3. INTAKE MOTORIZED DAMPERS SHALL BE INTERLOCKED WITH THEIR CORRESPONDING EXHAUST FANS:
   A. WHEN FANS ARE ENERGIZED, MOTORIZED DAMPERS SHALL OPEN.
   B. WHEN FANS ARE OFF, DAMPERS SHALL CLOSE.
   C. DAMPERS SHALL BE POWER TO OPEN, SPRING (FAIL) TO CLOSED POSITION.

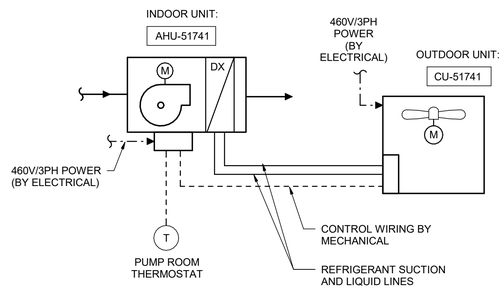
1 VENTILATION FANS - PUMP ROOM



CONTROL SEQUENCE:

1. THE SUPPLY FAN IS INTENDED TO PROVIDE TEMPERATURE RELIEF INSIDE THE OPERATOR ROOM.
2. PROVIDE A HAND-OFF-AUTO (H-O-A) SWITCH AT THE TEMPERATURE CONTROL PANEL FOR THE FAN.
   A. IN AUTO MODE THE FAN SHALL BE CONTROLLED BY THE THERMOSTAT. WHEN TEMPERATURE RISES ABOVE SETPOINT (85°F ADJUSTABLE), THE FAN SHALL START.
   B. WHEN ROOM TEMPERATURE IS BELOW SETPOINT, FAN SHALL REMAIN OFF AND ASSOCIATED MOTORIZED DAMPER SHALL REMAIN CLOSED.
   C. PROVIDE A 2°F DEADBAND TO PREVENT FREQUENT CYCLING.
   D. IN HAND MODE, THE FAN SHALL RUN CONTINUOUSLY.
3. THE MOTORIZED DAMPER SHALL BE INTERLOCKED WITH THE FAN OPERATION:
   A. WHEN THE FAN IS ENERGIZED, MOTORIZED DAMPER SHALL OPEN.
   B. WHEN THE FAN IS OFF, THE DAMPER SHALL CLOSE.
   C. DAMPERS SHALL BE POWER TO OPEN, SPRING (FAIL) TO CLOSE POSITION.

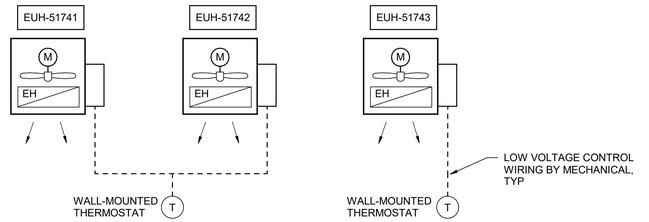
3 VENTILATION FAN - OPERATIONS ROOM



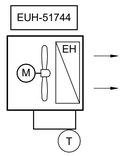
CONTROL SEQUENCE:

1. PROVIDE MANUFACTURER'S THERMOSTAT WITH 7-DAY PROGRAMMABLE OPERATING SCHEDULES.
2. FAN AND COMPRESSOR SHALL CYCLE ON/OFF AS REQUIRED TO MAINTAIN INDOOR TEMPERATURE.
3. MUTLI-STAGE COOLING:
   A. COMPRESSORS SHALL STAGE ON/OFF AS REQUIRED TO MAINTAIN INDOOR TEMPERATURE.
4. INDOOR TEMPERATURE SETPOINT SHALL BE AS FOLLOWS:
   A. COOLING SETPOINT: 80°F
   B. 24 HOURS A DAY, 7 DAYS A WEEK
5. SPLIT SYSTEM SHALL REMAIN OFF WHEN FREE COOLING CONDITIONS ARE MET AND EXHAUST FANS ARE ON AND MOTOR OPERATED DAMPERS ARE OPEN.

2 SPLIT SYSTEM AIR HANDLER AND CONDENSING UNIT



SUSPENDED UNIT HEATERS - PUMP ROOM



WALL MOUNTED UNIT HEATER - OPERATIONS ROOM

CONTROL SEQUENCES:

1. PROVIDE A 7-DAY PROGRAMMABLE WALL-MOUNTED THERMOSTAT FOR UNIT HEATERS, UNLESS OTHERWISE SPECIFIED TO BE PROVIDED WITH INTEGRAL THERMOSTAT MOUNTED TO THE UNIT HOUSING.
2. SPACES SHALL BE MAINTAINED AT SETPOINT (55°F, ADJUSTABLE) 24 HRS A DAY, 7 DAYS A WEEK. UNIT HEATER FAN AND ELECTRIC HEAT SHALL CYCLE ON/OFF AS REQUIRED TO MAINTAIN SPACE TEMPERATURE AT SETPOINT.
3. PROCESS ROOM:
   A. UNIT HEATERS IN THE PUMP ROOM ARE INTENDED TO BE ZONED AS INDICATED, SUCH THAT UNIT HEATERS EUH-51741 AND EUH-51742 ARE CONTROLLED BY A SINGLE THERMOSTAT. EUH-51743 IS INTENDED TO BE CONTROLLED BY A DEDICATED THERMOSTAT.

4 UNIT HEATERS

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A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: A. PUTMAN
DRAWN: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

SUB SHEET NO. 052 M701

TITLE OF SHEET
RSPH
HVAC CONTROLS

NORTH RIM - GRAND CANYON NATIONAL PARK

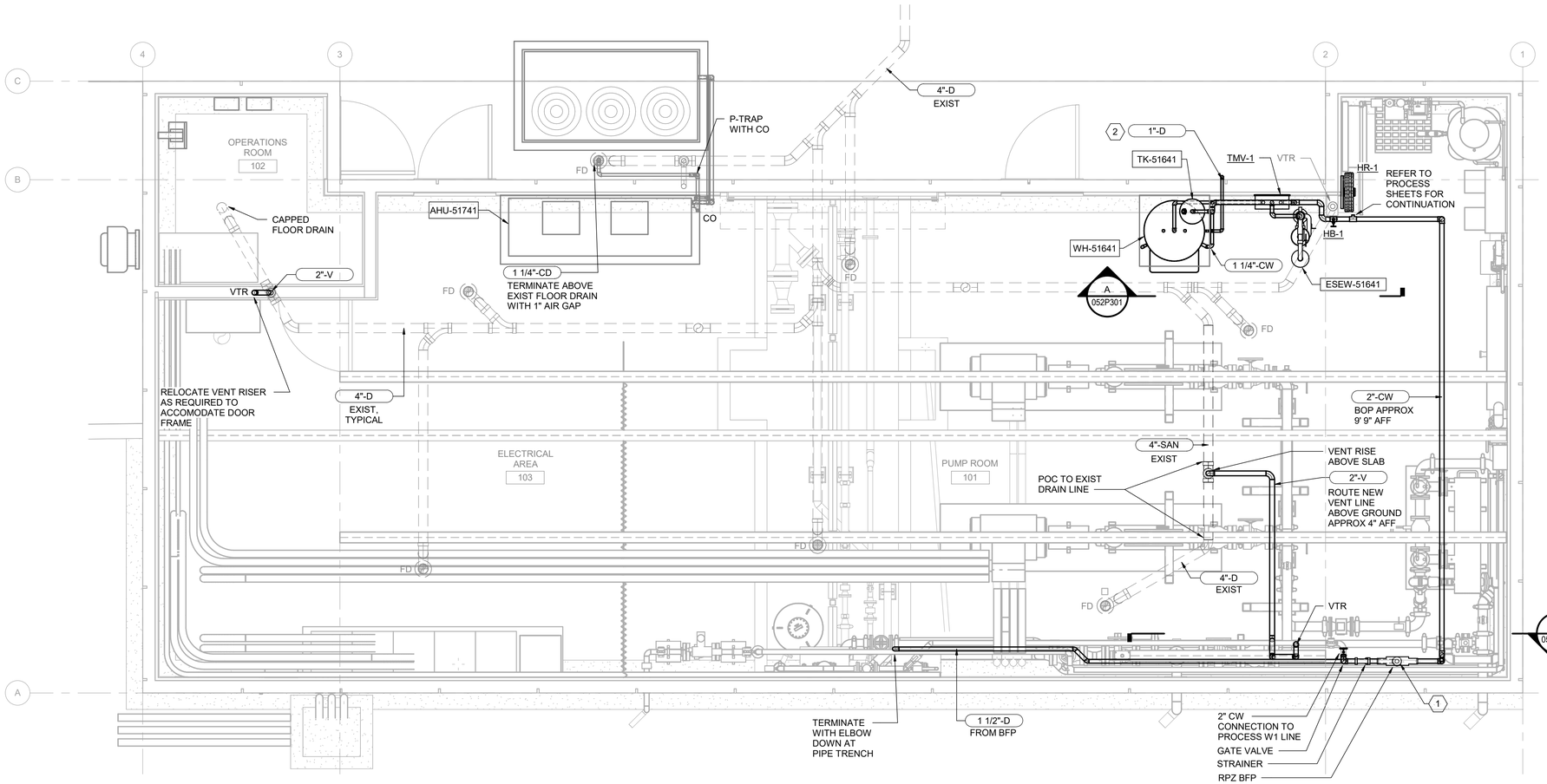
DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 84 OF 113

GENERAL NOTES:

A. REFER TO PROCESS DISCIPLINE 000 SERIES SHEETS FOR PIPE PENETRATION DETAILS.

KEYNOTES: (X)

1. PROVIDE DRAIN WITH AIR GAP FITTING FROM BFP, TERMINATE AT TRENCH.
2. DRAIN FROM WATER HEATER TEMPERATURE/ PRESURE RELIEF VALVE, DISCHARGE OUTSIDE, TERMINATE APPROX 4" ABOVE FINISHED GRADE.

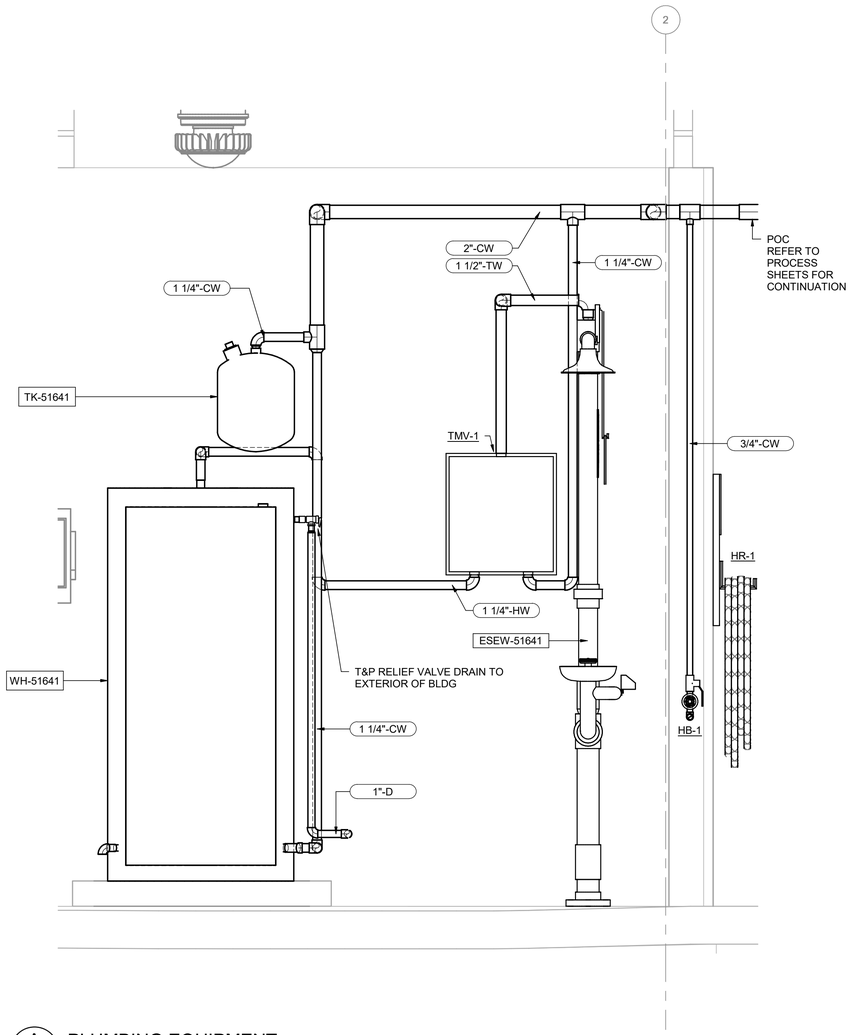


1 / 052P101 PLUMBING PLAN
SCALE (A)

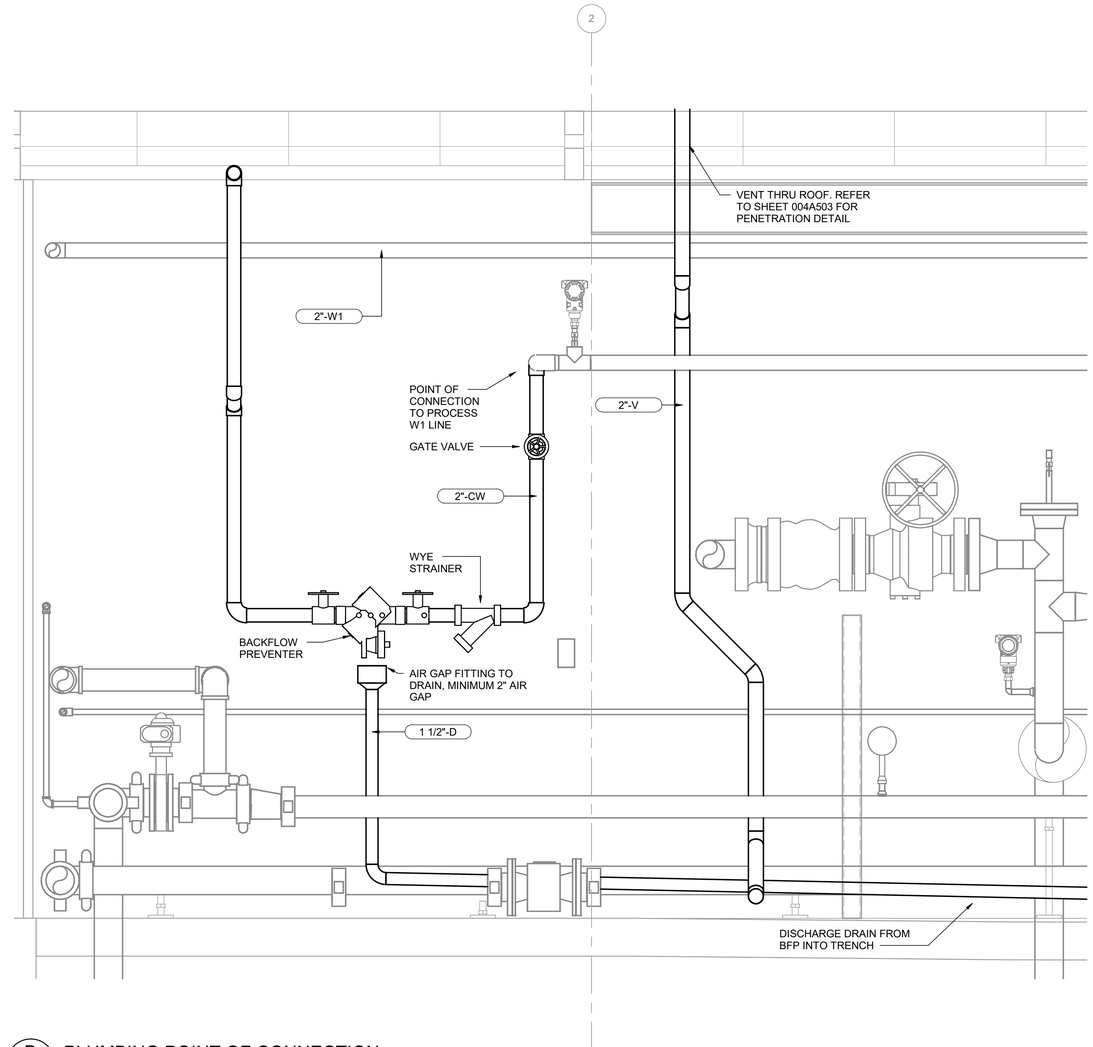
SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 052 P101	TITLE OF SHEET RSPH PLUMBING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 85 OF 113



A 052P101 PLUMBING EQUIPMENT
SCALE A

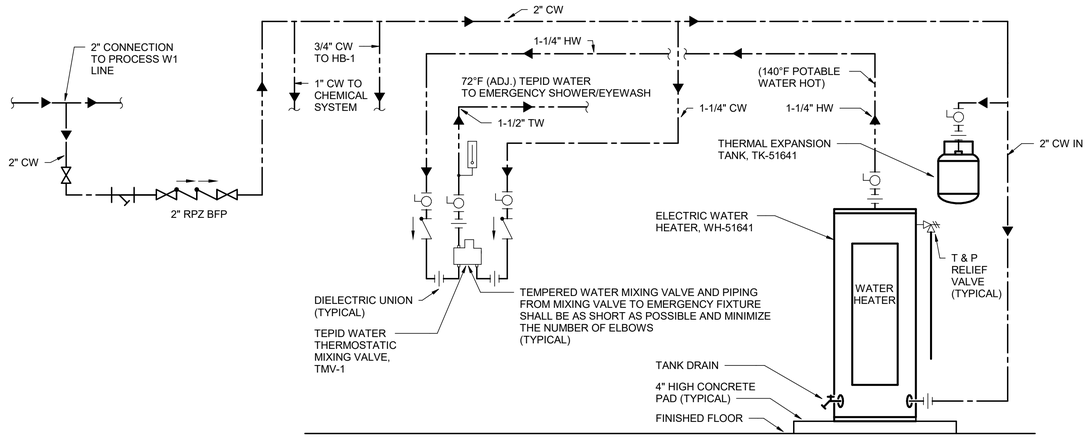


B 052P101 PLUMBING POINT OF CONNECTION
SCALE A

SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN CADD B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 052 P301	TITLE OF SHEET RSPH PLUMBING SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 86 OF 113

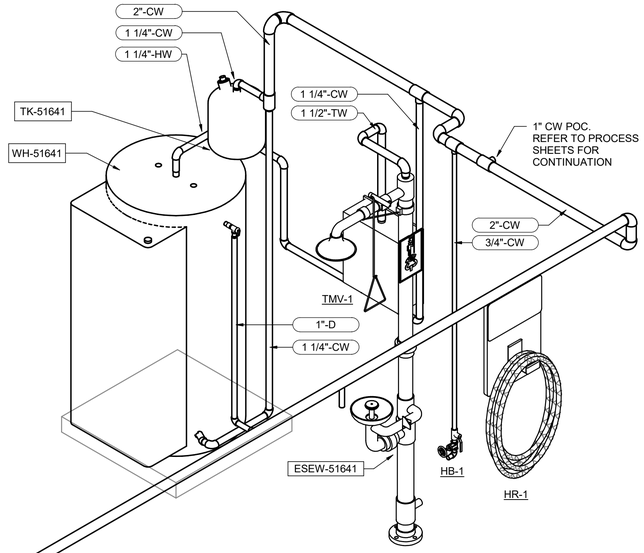
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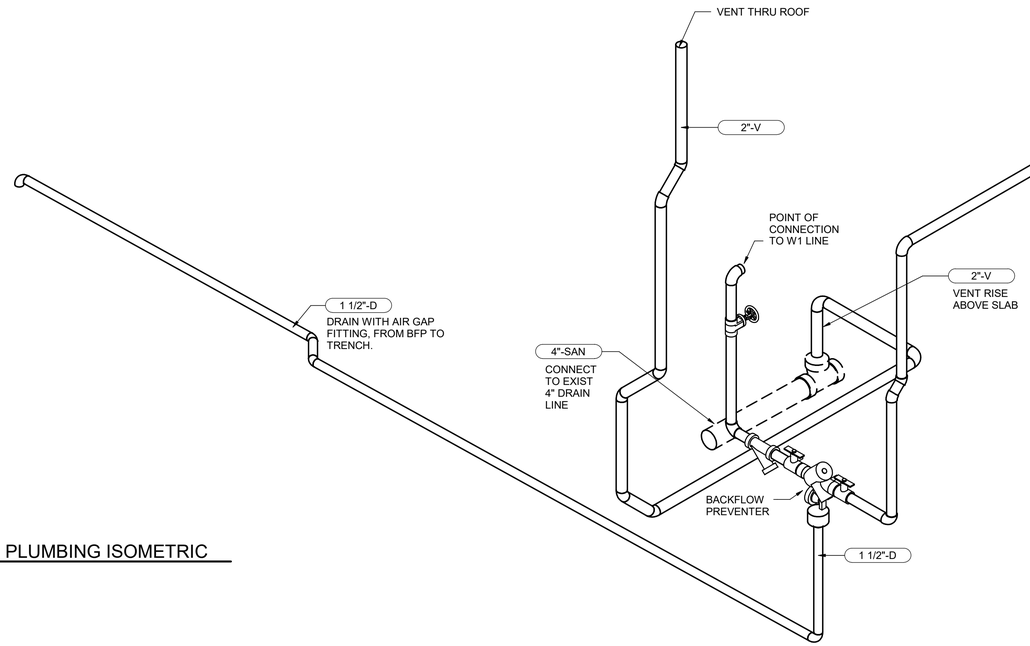


1 ELECTRIC WATER HEATER (WH-51641) PIPING DIAGRAM

GENERAL NOTES:

A. DRAWINGS ARE FOR SCHEMATICS PURPOSES ONLY. ALL VALVES & EQUIPMENT MUST BE INSTALLED PER
B. MANUFACTURER'S RECOMMENDATIONS FOR A FULLY FUNCTIONAL SYSTEM.
C. PROVIDE REQUIRED CLEARANCES TO COMBUSTIBLES AND SERVICING PER MANUFACTURER'S RECOMMENDATION.
D. PROVIDE REDUCERS AS NECESSARY FOR INSTALLATION OF EQUIPMENT.
E. PROVIDE THE MINIMUM CLEARANCES ON ALL SIDES PER MANUFACTURER'S RECOMMENDATIONS.
F. PROVIDE WATER HEATER WITH 20 GAUGE STEEL, 1" WIDE SEISMIC RESTRAINT STRAPS AT 1/3 AND 2/3 POSITION.
G. SECURE UNIT TO WALL WITH SCREWS.
H. ROUTE ALL DRAIN PIPES TO NEARBY FLOOR DRAIN.
I. INSTALL TEPID/TEMPERED WATER THERMOSTATIC MIXING VALVES PER MANUFACTURER'S RECOMMENDATIONS.
J. REFER TO STRUCTURAL PLANS FOR CONCRETE PAD DETAIL.





2 PLUMBING ISOMETRIC

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. PUTMAN DRAWN: B. BAXTER TECH. REVIEW: H. PEDRAZA DATE: MAY 2026	SUB SHEET NO. 052 P501	TITLE OF SHEET RSPH PLUMBING DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 87 OF 113

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DOMESTIC ELECTRIC WATER HEATERS

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	KW	ELECTRICAL VOLT/HZ/PH	RECOVERY RATE (GPH @ 90F RISE)	STORAGE CAPACITY (GAL)	WEIGHT (LBS)	NOTES
WH-51641	A.O. SMITH DRE-120-13.5	ROARING SPRINGS PUMPHOUSE	PUMP ROOM	13.5	480/60/3	62	119	400	1

NOTES:

1) FURNISH AND INSTALL STRAPS TO SEISMICALLY SECURE UNIT TO WALL.

TANK SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	TYPE	TANK VOLUME (GAL)	DIMENSIONS WIDTH (IN)	DIMENSIONS HEIGHT (IN)	WEIGHT (LBS)	NOTES
TK-51641	AMTROL ST-12-C-DD	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	EXPANSION, DIAPHRAGM	6.4	12	18	17	-

NOTES:

-

PLUMBING FIXTURE SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	BUILDING SERVED	AREA SERVED	FIXTURE TYPE	WASTE	VENT	COLD WATER	HOT WATER	NOTES
ESEW-51641	HAWS 8336	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	EMERGENCY SHOWER/ EYEWASH	-	-	-	-	-
HB-1	-	ROARING SPRINGS PUMP HOUSE	PUMP ROOM	HOSE BIB	-	-	3/4"	-	1

NOTES:

1) GLOBE VALVE WITH MALE THREADED HOSE ADAPTER AND VACUUM BREAKER

TEPID WATER MIXING VALVE SCHEDULE

TAG NO.	BASIS OF DESIGN MAKE/MODEL	APPLICATION	MINIMUM FLOW (GPM)	DESIGN FLOW (GPM)	DESIGN PRESSURE DROP (PSI)	RATED OUTLET TEMPERATURE RANGE (F)	OUTLET TEMPERATURE SETTING (°F)	CONNECTION SIZE (IN)	NOTES
TMV-1	HAWS TWBS.SHE	EMERGENCY EYE WASH / SHOWER	1	23	<5	60 - 90	75	1-1/4"	1,2,3

NOTES:

1) MIXING VALVE SHALL BE CERTIFIED PER ANSI Z358.1 FOR USE ON EMERGENCY FIXTURES.

2) PROVIDE SURFACE MOUNT STAINLESS STEEL CABINET.

3) MIXING VALVES SHALL COMPLY WITH NSF 61/372 FOR USE IN POTABLE WATER APPLICATIONS.

GENERAL NOTES:

A. PLUMBING SCHEDULES PRESENT INFORMATION ON THE SALIENT CHARACTERISTICS OF PLUMBING EQUIPMENT.

B. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. OR EQUAL SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS.

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_MP.rvt
4/27/2026 5:32:10 PM

A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED: A. PUTMAN
DRAWN: B. BAXTER
TECH. REVIEW: H. PEDRAZA
DATE: MAY 2026

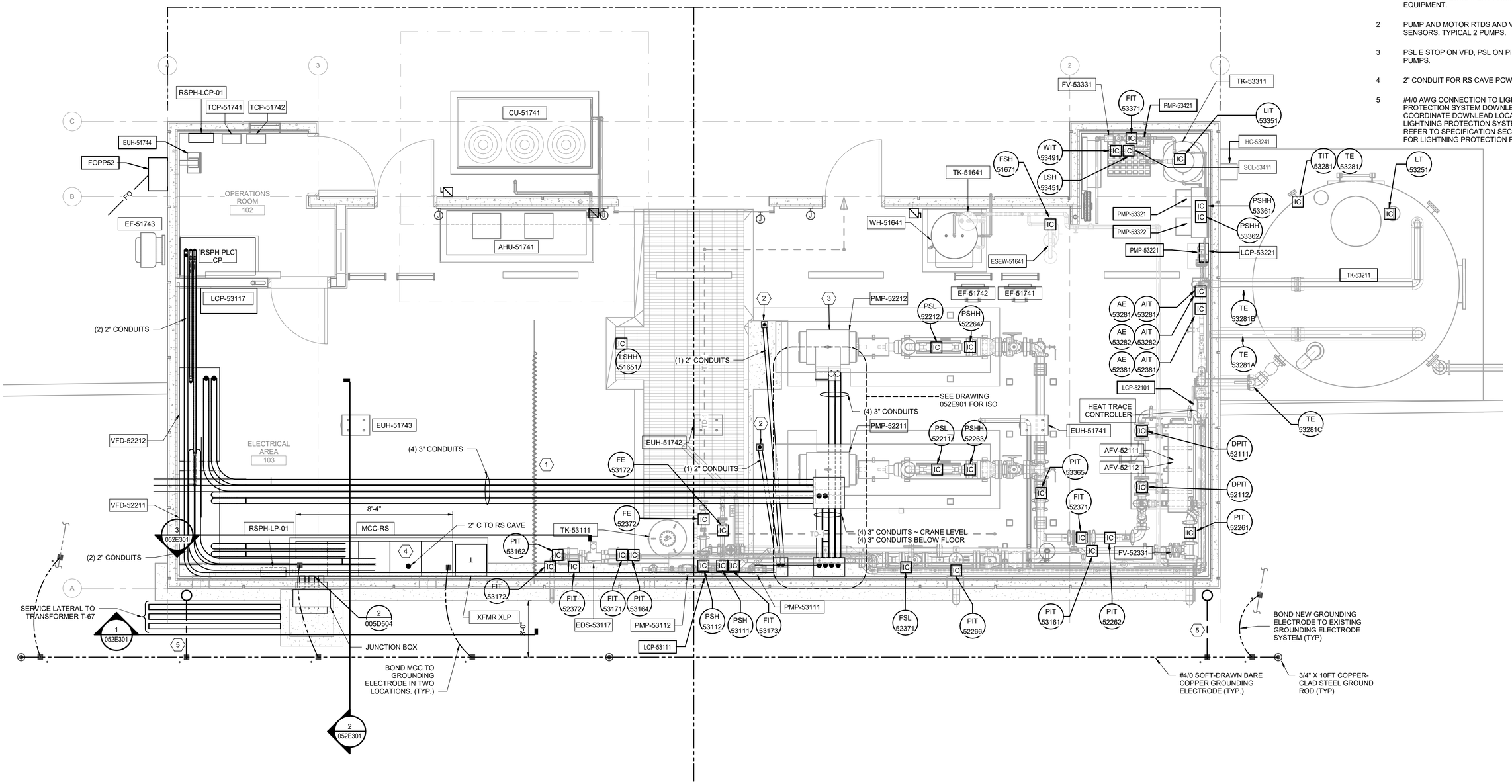
SUB SHEET NO. 052 P601

TITLE OF SHEET
RSPH
PLUMBING SCHEDULES
NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 88 OF 113

KEYNOTES (X)

1. REMOVABLE ROOM PARTITION TO ISOLATE ELECTRICAL EQUIPMENT FROM PUMP EQUIPMENT.
2. PUMP AND MOTOR RTDS AND VIBRATION SENSORS. TYPICAL 2 PUMPS.
3. PSL E STOP ON VFD, PSL ON PIPING. TYPICAL 2 PUMPS.
4. 2" CONDUIT FOR RS CAVE POWER.
5. #4/0 AWG CONNECTION TO LIGHTNING PROTECTION SYSTEM DOWNLEAD. COORDINATE DOWNLEAD LOCATIONS WITH LIGHTNING PROTECTION SYSTEM DESIGNER. REFER TO SPECIFICATION SECTION 26 41 13 FOR LIGHTNING PROTECTION REQUIREMENTS.



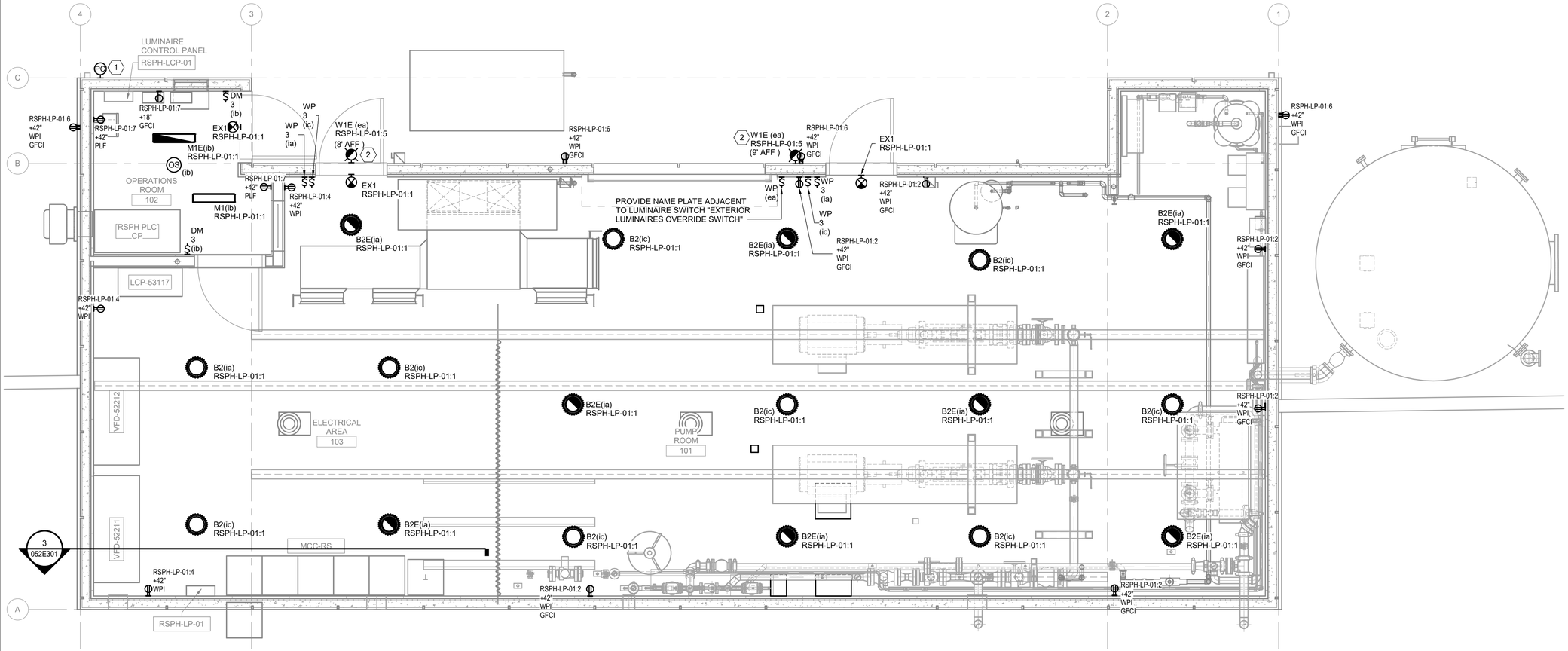
3 ELECTRICAL POWER PLAN
SCALE (A)

SCALE (A) 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 1/14/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE CADD: N. LYNCH TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E111	TITLE OF SHEET RSPH ELECTRICAL POWER PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 89 OF 113

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1 052E121 LIGHTING AND RECEPTACLE PLAN
SCALE A

GENERAL NOTES

A. REFER TO SHEET 052-E631 FOR LUMINAIRE SCHEDULES. REFER TO SHEET 008-E503 FOR LUMINAIRE DETAILS AND SHEET 008-E504 FOR LUMINAIRE SCHEMATICS.

B. COORDINATE FINAL LOCATION OF LUMINAIRES WITH OTHER TRADES PRIOR TO ROUGH-IN. ADJUST LUMINAIRE LOCATIONS AND HEIGHT ACCORDINGLY TO AVOID ANY CONFLICTS.

C. COORDINATE LOCATION OF LUMINAIRES WITH STRUCTURE. PROVIDE ADDITIONAL STRUCTURAL SUPPORTS FOR LUMINAIRES IN ORDER TO ACHIEVE REQUIRED MOUNTING HEIGHT.

D. EMERGENCY EXIT SIGNS AND EMERGENCY LUMINAIRE UNITS TO BE UNSWITCHED.

E. REFER TO POWER PLANS FOR AREA CLASSIFICATION TYPES.

F. REFER TO LUMINAIRE CONTROLS SEQUENCE SCHEDULE ON SHEET 052-E631 FOR ALL INTERIOR AND EXTERIOR SPACE/AREA TYPE LUMINAIRE CONTROL REQUIREMENTS.

KEYNOTES (X)

1 PHOTOCELL FOR EXTERIOR BUILDING LUMINAIRE CONTROLS 'RSPH-LCP-01'. MOUNT AT 9'-0" AFF, FACING NORTH. REFER TO DETAIL #2 ON SHEET 008-E504.

2 EXTERIOR LUMINAIRE CONTROLLED BY LUMINAIRE CONTROL RELAY PANEL 'RSPH-LCP-01' AND LOCAL LOW-VOLTAGE OVERRIDE SWITCH DEVICE.

PHOTOMETRIC AND POWER DENSITY STATISTICS - INTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LIGHT LOSS FACTOR (LLF)	W/FT² (WHOLE BLDG)
101 - PUMP ROOM	40 FC @ FF	3.0:1	0.82	0.60
102 - SCADA / IT CLOSET	40 FC @ FF	2.0:1	0.90	
103 - ELECTRICAL ROOM	40 FC @ 30" AFF	3.0:1	0.82	

NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

PHOTOMETRIC AND POWER DENSITY STATISTICS - EXTERIOR

DESCRIPTION	AVERAGE	MAX./MIN.	LLF	W/FT2	TOTAL LINEAR FT.	W/LINEAR FT.
BUILDING ENTRY DOORS	1 FC @ FF	5.0:1	0.78	N/A	6	11.67
LOADING DOCK	1 FC @ FF	5.0:1	0.78	0.02	N/A	N/A

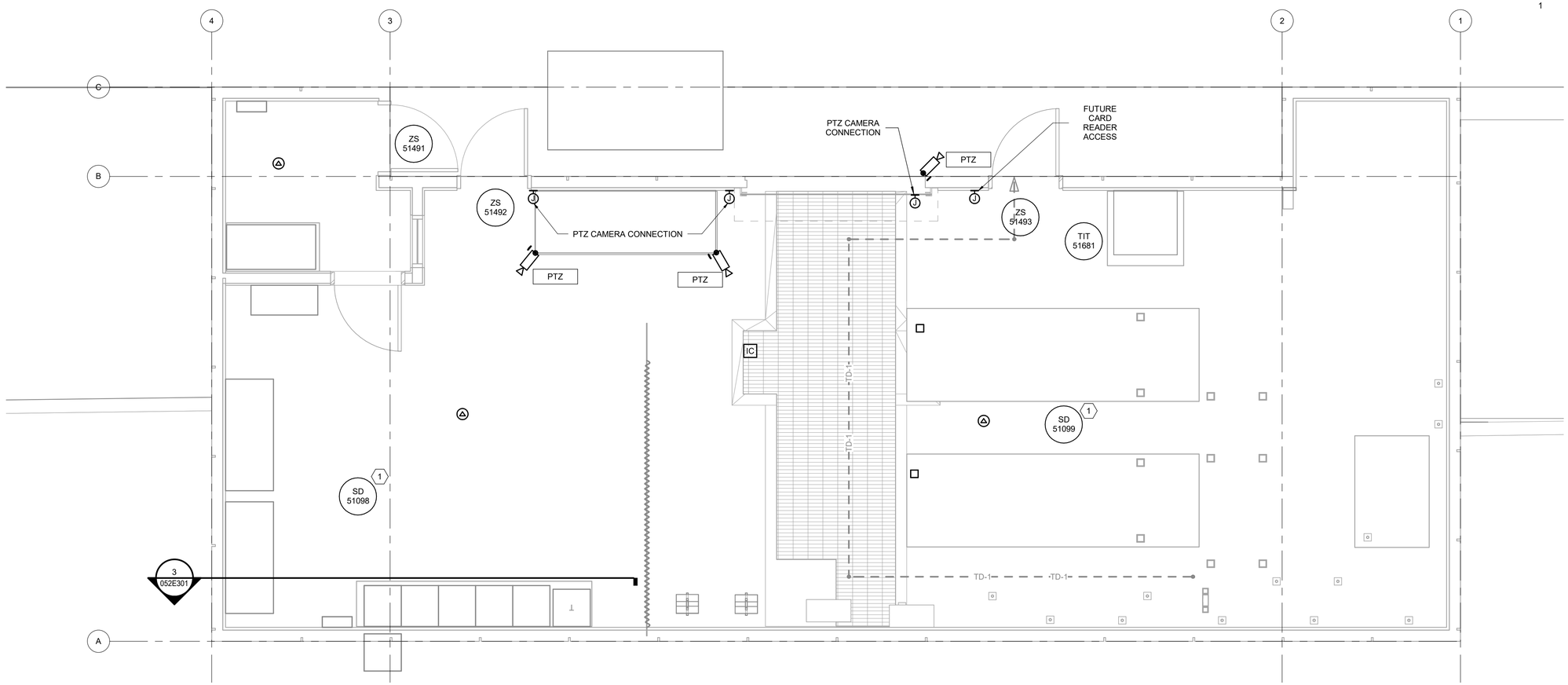
NOTE:
1. CONTRACTOR TO SUBMIT WHOLE BUILDING (INDIVIDUAL INTERIOR ROOMS/ SPACES) AND PROJECT LIMITS (EXTERIOR BASE SITE, PARKING, ENTRANCES/ EXITS, CANOPIES, WALKWAYS, AND BUILDING FAÇADE) PHOTOMETRIC AND POWER DENSITY ANALYSIS FOR ALL SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS NOT LISTED ON THE LUMINAIRE SCHEDULE. SUBMITTED APPROVED EQUAL/ ALTERNATE LUMINAIRE MANUFACTURERS AND MODEL NUMBERS MUST MEET THE REQUIRED PHOTOMETRIC AND POWER DENSITY STATISTICS FOR ALL AREAS/ SPACES INDICATED ON THIS SHEET. INCREASING LUMINAIRE QUANTITIES SHOWN ON THIS SHEET, INCREASING THE LUMEN OUTPUT ON THE LUMINAIRE SCHEDULE, OR INCREASING THE VA (VOLTAGE AMPS) ON THE LUMINAIRE SCHEDULE FOR THE PHOTOMETRIC AND POWER DENSITY STATISTICS COMPLIANCE IS PROHIBITED.

SCALE A 2 0 2 4 6 3/8" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL CERTIFICATE NO. 58037 TRAVIS T. MOORE Date signed 1/14/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. MARTINEZ DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E121	TITLE OF SHEET RSPH LIGHTING AND RECEPTACLE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 90 OF 113

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1 SPECIAL SYSTEMS
SCALE A

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SCALE OF FEET

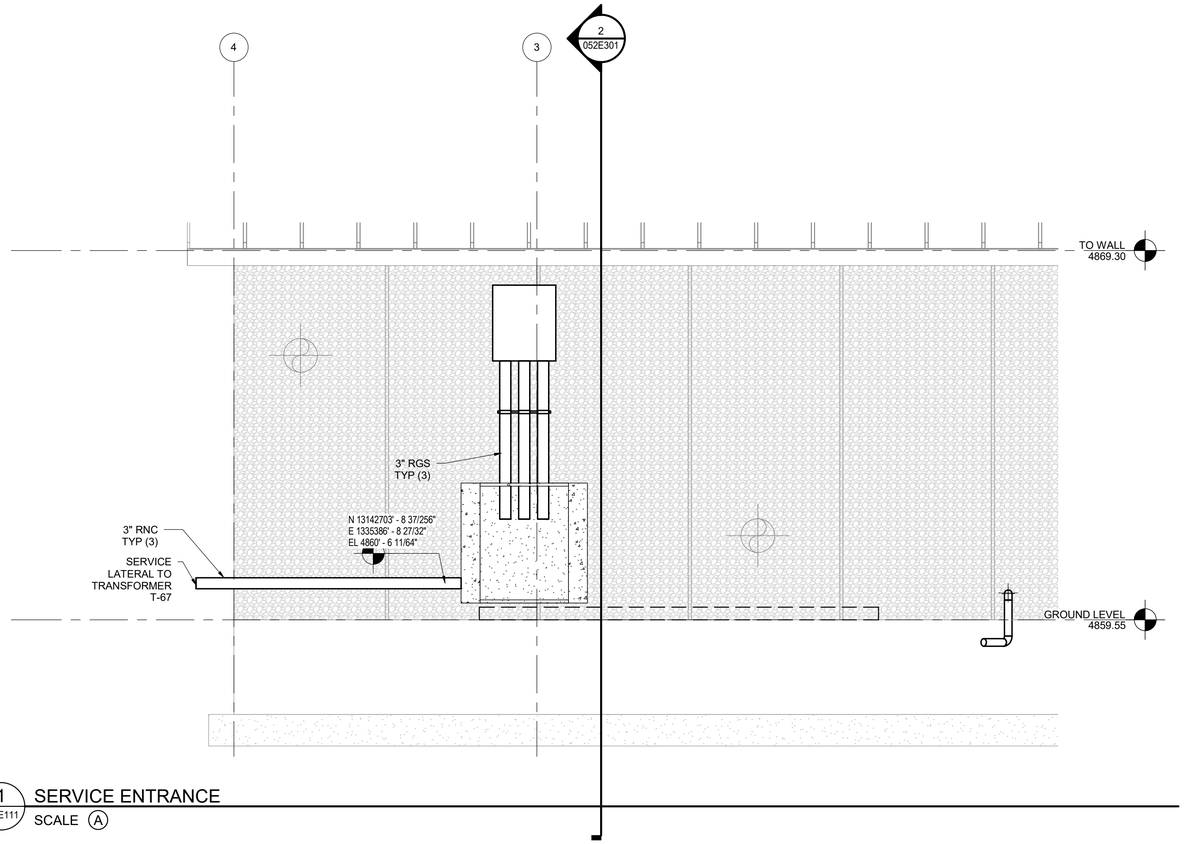
KEYNOTES X

1 SMOKE DETECTORS ARE FOR ASSET PROTECTION ONLY. NOT REQUIRED BY CODE. PROVIDE A PHOTOELECTRIC SMOKE ALARM WITH HORN/STROBE OUTPUT AND FORM C RELAY CONTACT OUTPUT. CONTRACTOR TO PROVIDE 120VAC.

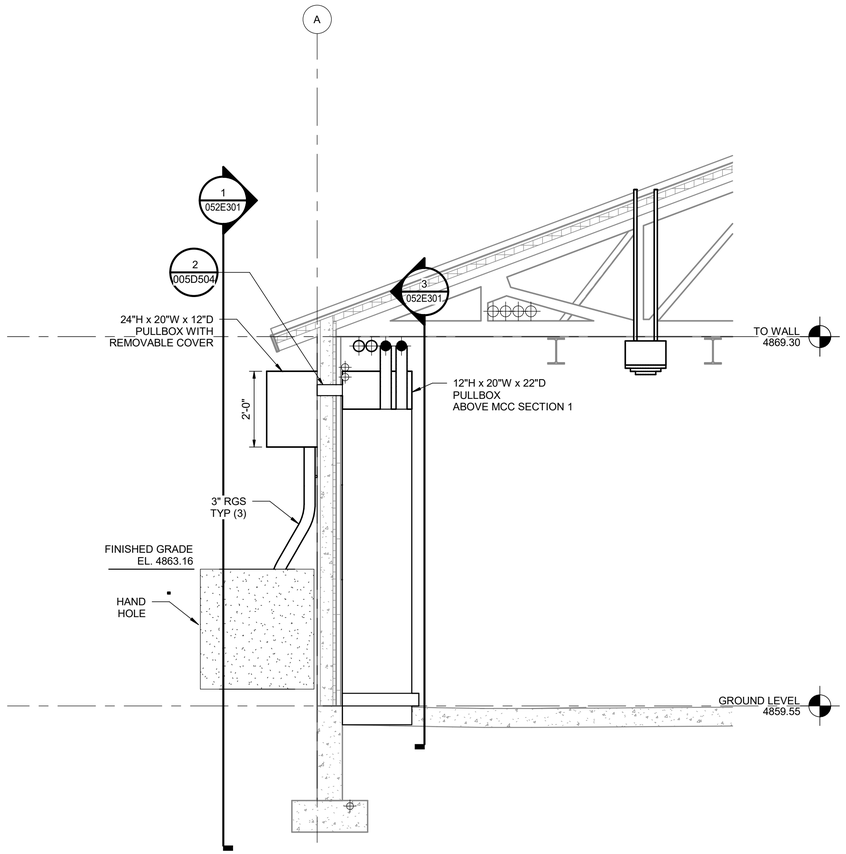
REGISTERED PROFESSIONAL ENGINEER (DEPARTMENT) CERTIFICATE NO. 58037 TRAVIS T. MOORE Date Signed 1/AUG/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAFTED D. LOPEZ TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E131	TITLE OF SHEET RSPH SPECIAL SYSTEMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 91 OF 113

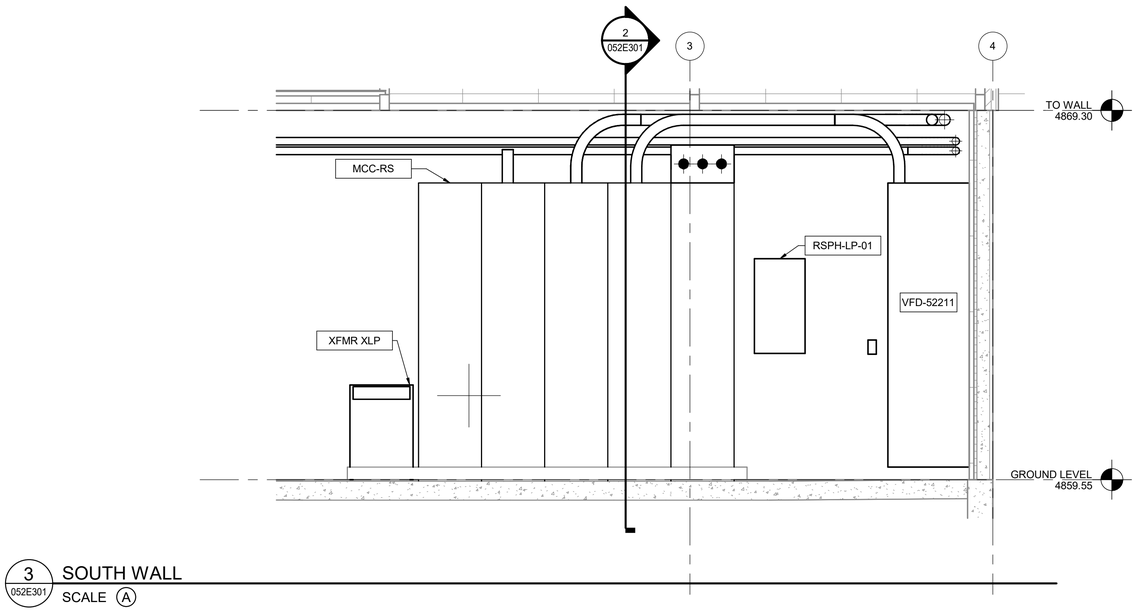
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1 052E111 SERVICE ENTRANCE
SCALE (A)



2 052E111 SECTION 7
SCALE (A)



3 052E301 SOUTH WALL
SCALE (A)

SCALE (A) 2 0 2 4 1/2" = 1'-0"
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 10/10/2025
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: S.BILD TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E301	TITLE OF SHEET SERVICE ENTRANCE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 92 OF 113

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MCC LOAD ANALYSIS:	TAG	UNIT	LOAD	AMPS @ 480V
LIGHTING XFMR XLP	MCC1-4A	KVA	45	54
DIRECT CONNECTIONS				
VFD-52211	MCC1-2E	HP	450	515
VFD-52212 (STANDBY)	MCC1-3E	HP	450	0
AHU-51741	MCC1-4CL	HP	4	4.8
HVAC CONDENSING UNIT CU-51741	MCC1-4CR	KW	20.7	24.9
UNIT HEATER EUH-51741	MCC1-4EL	KW	3.7	4.5
UNIT HEATER EUH-51742	MCC1-4ER	KW	3.7	4.5
UNIT HEATER EUH-51743	MCC1-4GL	KW	3.7	4.5
RW BYPASS VALVE FCV-52331	MCC1-JL	HP	1	1.2
FW TANK HEATER HTR-53241	MCC1-4JR	KW	2.6	3.1
ESEW WATER HEATER WH-51641	MCC1-4LL	KW	13.5	16.2
FEEDER TO ROARING SPRINGS CAVE	MCC1-4LL	KW	10	20.8
FV-33331 - CHLORINE XFER MAKEUP WATER VALVE	MCC1-5CL	HP	1/2	1.1
SUBTOTAL MCC LOAD				**654.6**
25% OF LARGEST MOTOR (VFD-52211)				**128.8**
25% SPARE AMPACITY PER NPS				**195.8**
SUBTOTAL MCC MINIMUM AMPACITY				**979.1**

GENERAL NOTES

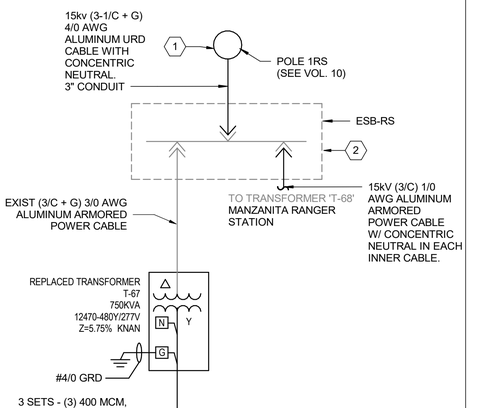
A. ROARING SPRINGS PUMP #1 AND PUMP #2 ARE ELECTRICALLY-INTERLOCKED AND SOFTWARE-INTERLOCKED TO ALLOW ONLY ONE PUMP TO OPERATE AT ANY TIME.

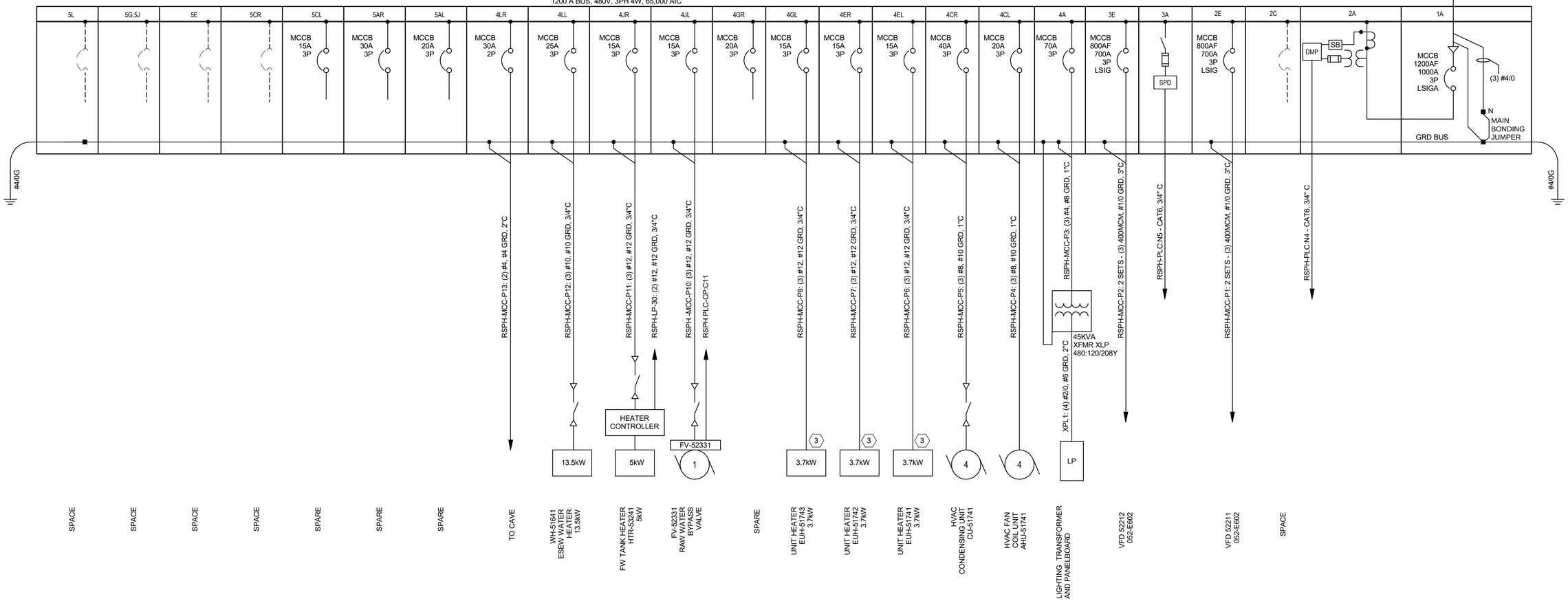
KEYNOTES (X)

1 OVERHEAD LINE TO NORTH RIM.

2 ESB-RS THREE JUNCTION ENCLOSURES WITH LOAD BREAK ELBOW TERMINATIONS AT BOTTOM OF OVERHEAD POWER LINE.

3 ALL ELECTRIC UNIT HEATERS WILL BE FURNISHED WITH MANUFACTURER SUPPLIED INTERNAL DISCONNECT SWITCHES.

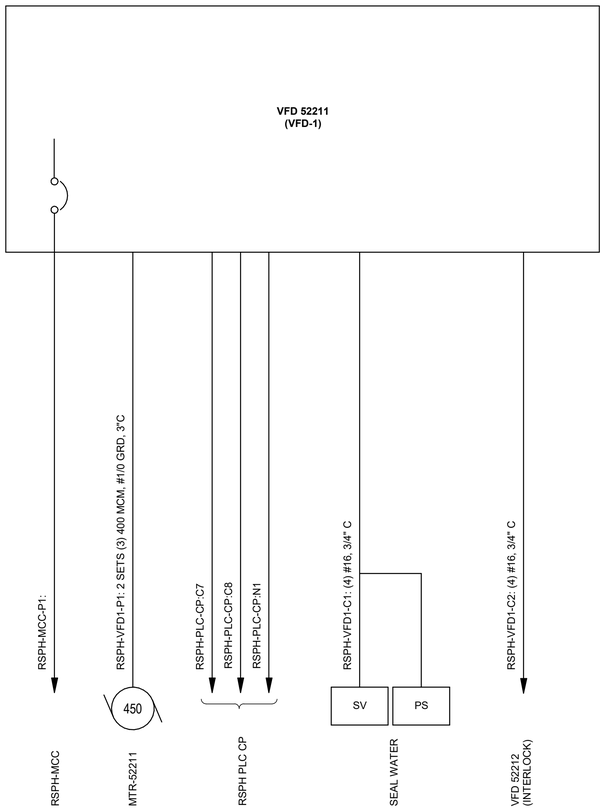




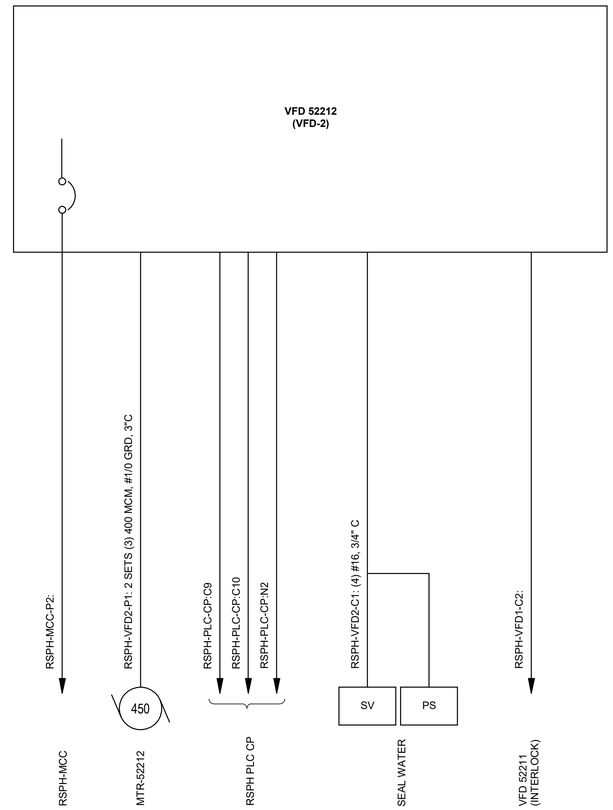
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REGISTERED PROFESSIONAL ENGINEER — 58037 — TRAVIS T. MOORE — ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: S.BILD TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E601	TITLE OF SHEET RSPH ONE-LINE DIAGRAM MCC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 93 OF 113



VFD-52211 ONE-LINE DIAGRAM



VFD-52212 ONE-LINE DIAGRAM

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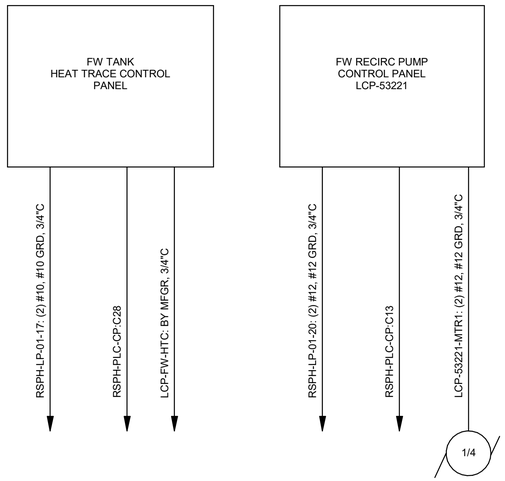
REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date 4/27/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER DRAWN: S.BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E602	TITLE OF SHEET RSPH VFD 52211 AND VFD 52212 ONE-LINE DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 94 OF 113

KEYNOTES: (X)

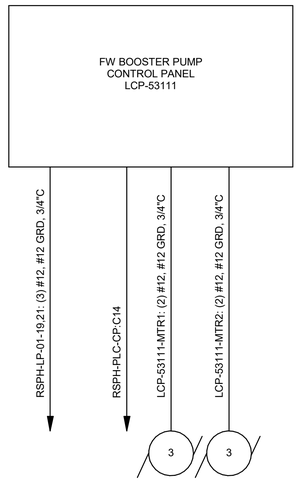
1. TEMPERATURE CONTROL PANELS PROVIDED UNDER MECHANICAL.

2. WIRING AND CONDUIT TO TEMPERATURE CONTROL ACCESSORIES PROVIDED UNDER MECHANICAL.

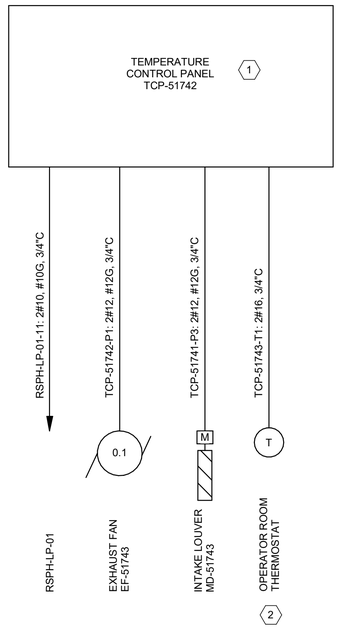


FW TANK HEAT TRACE
CONTROL PANEL LCP-FW-HTC
ONE-LINE DIAGRAM

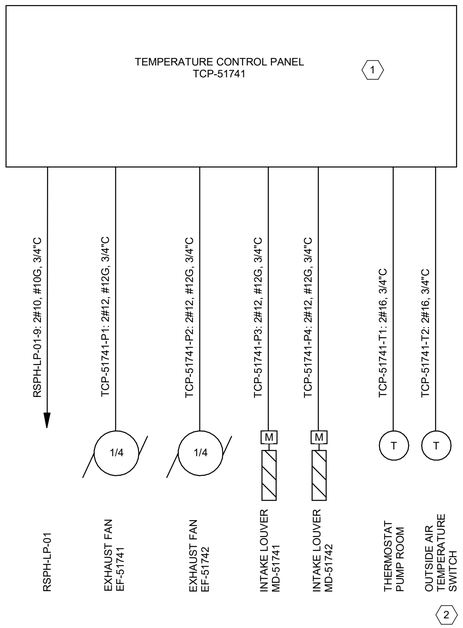
FW RECIRC PUMP
CONTROL PANEL LCP-53221
ONE-LINE DIAGRAM



FW BOOSTER PUMPS
CONTROL PANEL LCP-53111
ONE-LINE DIAGRAM



TEMPERATURE
CONTROL PANEL TCP-51742
ONE-LINE DIAGRAM

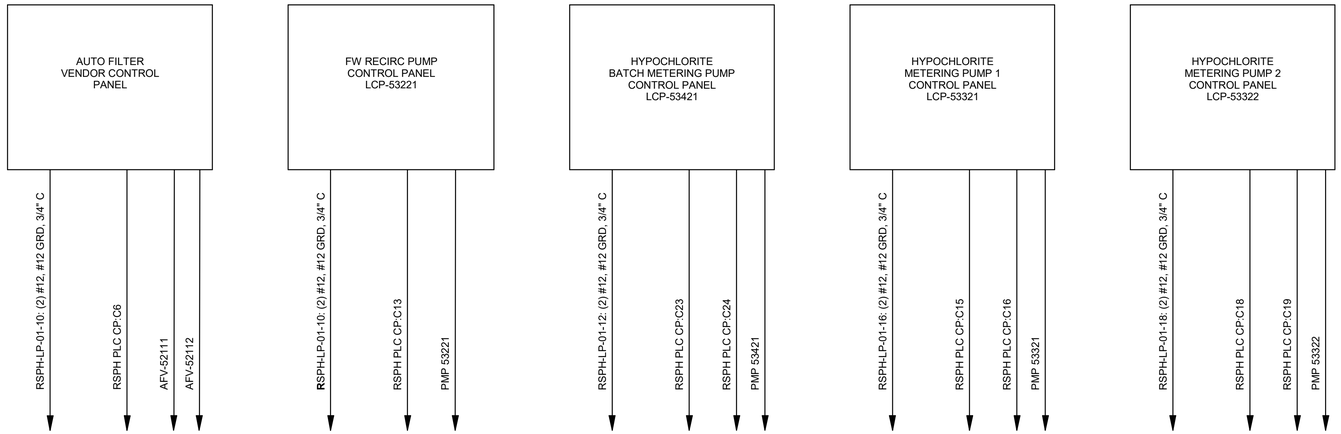


TEMPERATURE CONTROL PANEL TCP-51741 ONE-LINE DIAGRAM

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REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO.
58037
TRAVIS T.
MOORE
Date signed 10/12/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER DRAWN: S.BILD TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E603	TITLE OF SHEET RSPH MISCELLANEOUS ONE-LINE DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 95 OF 113



AUTOFLUSH FILTER
CONTROL PANEL LCP-52101
ONE-LINE DIAGRAM

FW RECIRC PUMP
CONTROL PANEL LCP-53221
ONE-LINE DIAGRAM

HYPOCHLORITE BATCH
METERING PUMP
CONTROL PANEL LCP-53421
ONE-LINE DIAGRAM

HYPOCHLORITE
METERING PUMP 1
CONTROL PANEL LCP-53321
ONE-LINE DIAGRAM

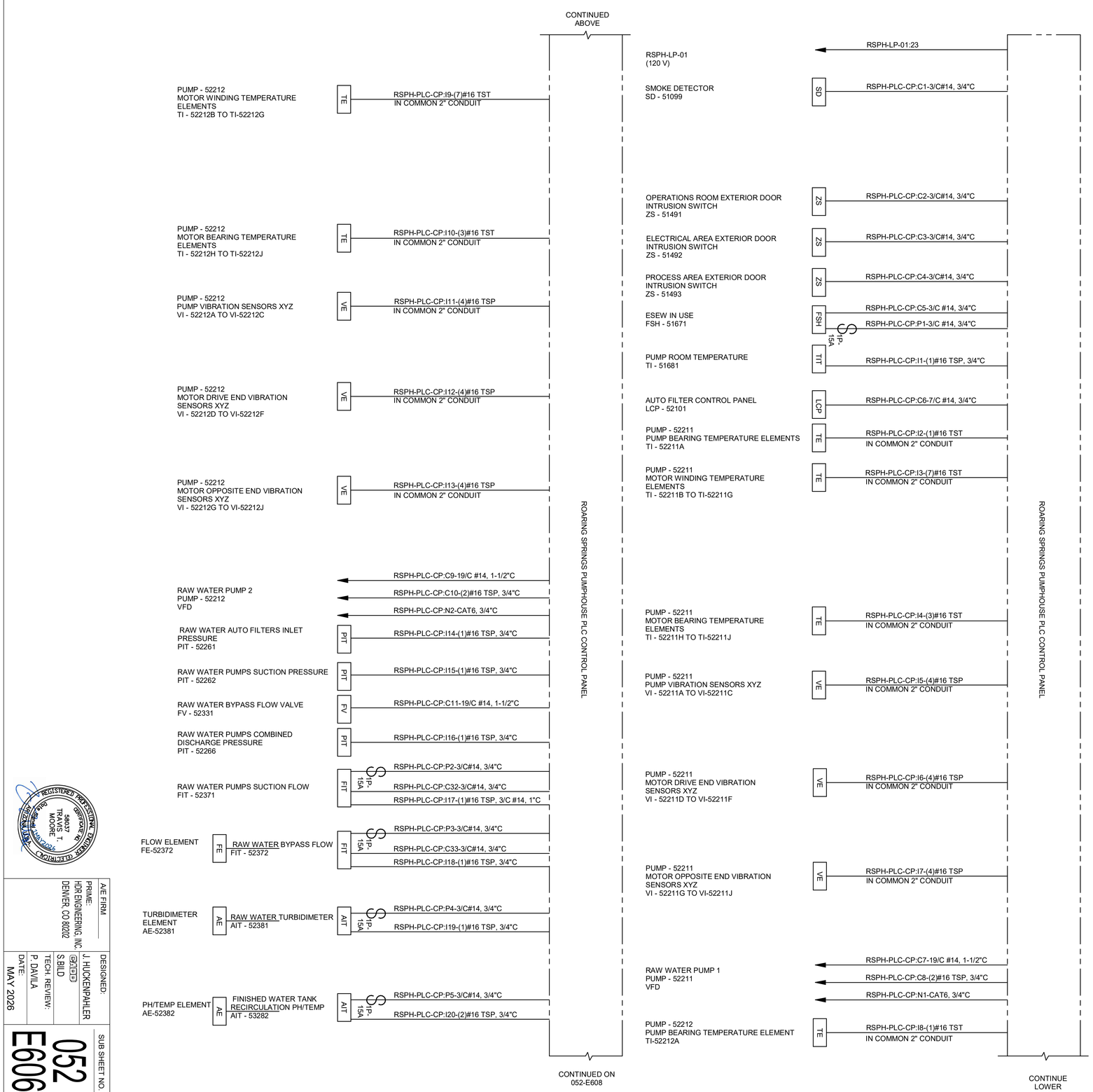
HYPOCHLORITE
METERING PUMP 2
CONTROL PANEL
LCP-53322
ONE LINE DIAGRAM

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO.
58037
TRAVIS T.
MOORE
Date Signed 1/31/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER DRAWN: S.BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E604	TITLE OF SHEET RSPH MISCELLANEOUS ONE-LINE DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 96 OF 113

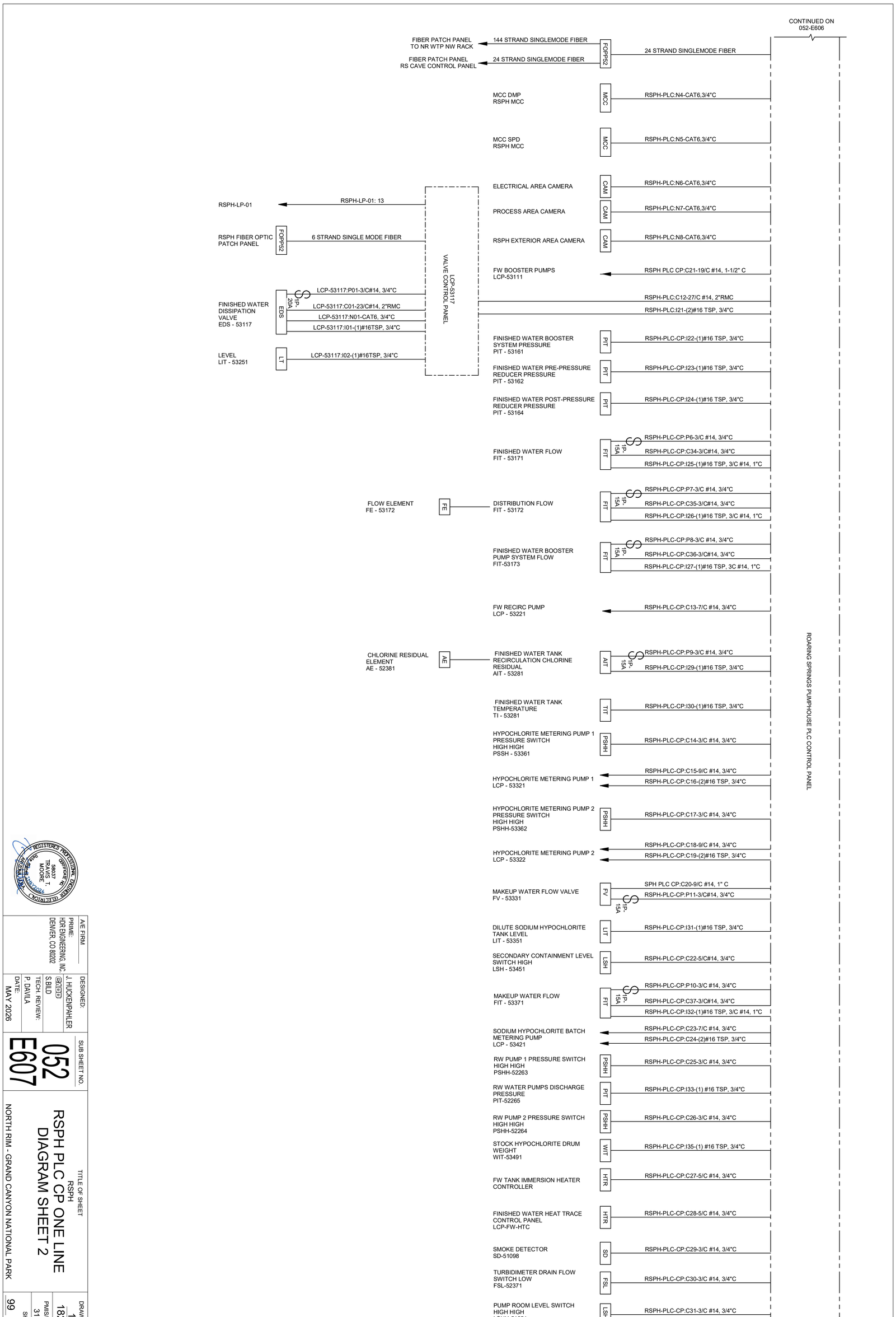
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REGISTERED PROFESSIONAL ENGINEER
58037
TRAVIS T. MOORE
ELECTRONICALLY SIGNED AND SEALED

A/E FIRM	
PRIME:	HDR ENGINEERING, INC.
	DENVER, CO 80202
DESIGNED:	J. HUCKENPAHLER
DRAWN:	S.BILD
TECH. REVIEW:	P. DAVILA
DATE:	MAY 2026
SUB SHEET NO.	052 E606
TITLE OF SHEET	RSPH PLC CP ONE LINE DIAGRAM SHEET 1
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	98 OF 113



A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202
DESIGNED: J. HUCKENPAHLER
DRAWN: S.BILD
TECH. REVIEW: P. DAVILA
DATE: MAY 2026

SUB SHEET NO. E607 052

TITLE OF SHEET
RSPH
RSPH PLC CP ONE LINE DIAGRAM SHEET 2

NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 99 OF 113

LUMINAIRE SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	LAMPS							MOUNTING	HEIGHT	DESCRIPTION	IMAGE	NOTES
		TYPE	TEMP	LUMENS	CRI	BUG	VOLTAGE	VA					
B2/ B2E	HOLOPHANE "PETROLUX" #PXLW-5000LM-MDU5FR-120-40K-80CRI-PM-nPP16D-WGX-CR-DGXD WALL/STANCHION MOUNT OPTION: #P3US-DGXD WIREGUARD ACCESSORY: #WGPXLW-CR 'B2E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WMCP OR APPROVED EQUAL	LED	4000K	4403	80	N/A	120	35	PENDANT OR WALL (TO BOTTOM OF FIXTURE)	9'-9" AFF (OR AS NOTED PER DRAWINGS)	COMPACT HEAVY-DUTY INDUSTRIAL LUMINAIRE WITH ROBUST CAST ALUMINUM ALLOY (0.6%CU CONTENT) HOUSING, STAINLESS STEEL TAMPER-RESISTANT HARDWARE, DISTORTION RESISTANT HOLLOW CORE SILICONE TYPE GASKET, GRAY CORROSION RESISTANT POLYESTER POWDER COAT FINISH, ENDURAL PRISMATIC BOROSILICATE GLASS REFRACTORS WITH UNIFORMITY AND GLARE CONTROL, MEDIUM FROSTED W/ UPLIGHT DISTRIBUTION OPTICS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, INTEGRAL nLIGHT CONTROL MODULE, HIGH OUTPUT LONG-LIFE LEDs, HIGH EFFICIENCY DRIVER, INTEGRAL OPTIC WIREGUARD, INTEGRAL 90 DEGREE UNIVERSAL MOUNTING BRACKET FOR WALL OR 1-1/2" NPT STANCHION MOUNTING, INTEGRAL 3/4" NPT PENDANT MOUNTING, WET LOCATION RATED.		
EX1	LITHONIA #WLTE-W-(NOTE 1)-R-EL-TP-SD OR APPROVED EQUAL	LED	-	-	-	N/A	120/277	3	SURFACE, WALL, END, OR PENDANT (TO BOTTOM OF FIXTURE)	1'-0" ABOVE EGRESS DOOR FRAME/ CEILING (OR AS NOTED PER DRAWINGS)	INDUSTRIAL EXIT SIGN WITH IMPACT-RESISTANT SCRATCH-RESISTANT CORROSION-PROOF THERMOPLASTIC WHITE COMPACT HOUSING, RUGGED HEAVY-DUTY POLYCARBONATE CLEAR HOUSING LENS, TAMPER-RESISTANT SCREW HARDWARE, 6" HIGH WITH 3/4" STROKE RED LETTERS, UNIVERSAL REMOVABLE DIRECTIONAL CHEVRONS INDICATORS, INTEGRAL SEALED MAINTENANCE FREE NICKEL-CADMIUM EMERGENCY BATTERY PACK, INTEGRAL DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL SELF-DIAGNOSTIC TESTING, INTEGRAL UNIVERSAL (TOP, END, OR BACK) CANOPY MOUNT, WET LOCATION RATED.		1
M1/ M1E	HOLOPHANE #EMX-L24-6000LM-FGFL-WD-MVOLT-GZ10-40K-80CRI-SPD20KV-RMBK-DGXD-CR WALL ANGLE MOUNT: #EMXRMBA MONOPOINT PENDANT MOUNT: PMP4X PROVIDE nLIGHT POWER PACK 'PP' (SEE LUMINAIRE CONTROLS SCHEDULE) 'M1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E10WLCP OR APPROVED EQUAL	LED	4000K	4697	80	N/A	MVOLT	40	PENDANT, SURFACE, OR WALL (TO BOTTOM OF FIXTURE)	9' AFF OR CEILING STRUCTURE (OR AS NOTED PER DRAWINGS)	2' LOW-PROFILE LENSED INDUSTRIAL STRIP LUMINAIRE WITH COMPACT-DESIGN CAST & EXTRUDED ALUMINUM HOUSING & COVER, 18-8 STAINLESS STEEL HARDWARE, INTEGRAL MOLDED SILICON GASKET, 3/4" THREADED NPT END-CAPS, ENHANCED CORROSION RESISTANT DURABLE GREY POWDER COAT FINISH, 0.125" THICK FLAT FROSTED GLASS LENS, WIDE DISTRIBUTION OPTICS, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, INTEGRAL 0-10V DIMMING CONTROLS, iNTEGRAL nLIGHT CONTROL MODULE, SECONDARY OCCUPANCY/MOTION SENSOR CAPABILITY, HIGH OUTPUT LONG-LIFE LEDs, HIGH-EFFICIENCY DRIVER, INTEGRAL RIGID MOUNTING BRACKET, WALL ANGLED (22.5°,45°,90°) RIGID MOUNTING BRACKET OPTION, MONOPOINT PENDANT MOUNTING BRACKET OPTION, IP67/ WET/ NEMA 4X LOCATION RATED.		
W1/ W1E	LITHONIA "D-SERIES" #DSXW1-LED-P1-AMBLW-AMCRI-T2S-MVOLT-SRW-PIR1FC3V-VG-DDBXD 'W1E' - PROVIDE WITH INTEGRAL EMERGENCY BATTERY BACK-UP: #E20WC OR APPROVED EQUAL	LED	TRUE AMBER PHOSPHOR (595nm)	613	-	B0-U0-G0	MVOLT	10	WALL (TO BOTTOM OF LUMINAIRE)	A.F.F. AS NOTED PER DRAWINGS	SMALL EXTERIOR WALL PACK LUMINAIRE WITH TWO-PIECE DIE-CAST ALUMINUM HOUSING, INTEGRAL HEAT SINKS FINS, ONE-PIECE SOILD SILICONE GASKET, DARK BRONZE ZINC-INFUSED TGIC THERMOSET POWDER COAT FINISH, FULL "CUT-OFF" (NIGHTTIME FRIENDLY) VISUAL COMFORT TYPE II SHORT DISTRIBUTION OPTICS, INTEGRAL PRECISION-MOLDED ACRYLIC LENSES, INTEGRAL DRIVER DISCONNECT, INTEGRAL SURGE PROTECTION, HIGH OUTPUT LONG-LIFE LEDs, HIGH EFFICIENCY DRIVER, INTEGRAL LENS VANDAL GUARD, INTEGRAL MOTION/AMBIENT SENSOR ENABLED AT 1 FC, WET LOCATION RATED.		2

NOTES:

1) REFER TO DRAWINGS FOR EXACT NUMBER OF FACES REQUIRED AT EACH EXIT SIGN LOCATION. FIELD ADJUST DIRECTIONAL CHEVRON ACCORDINGLY FOR PATH OF EGRESS.

2) LUMINAIRES TO BE CONTROLLED VIA THE FACILITY PHOTOCELL/ TIME CLOCK/ RELAY-CONTACTOR LIGHTING CONTROL PANEL. REFER TO LIGHTING CONTROL SCHEMATIC DETAIL #2 ON SHEET 008-E504.

LUMINAIRE CONTROLS DEVICE SCHEDULE

TAG	MANUFACTURER/ MODEL NUMBER	VOLTAGE	MOUNTING	HEIGHT	DESCRIPTION	IMAGE	NOTES
OS	ACUITY nLIGHT #NCM-PDT-9-RJB-AR-20M-LT OR APPROVED EQUAL	CAT-5e	SURFACE	CEILING	DUAL TECHNOLOGY (MICROPHONICS AND DIGITAL PASSIVE INFRARED) CEILING/SURFACE MOUNTED OCCUPANCY SENSOR WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, SNAP-LOCKS COVER PLATE, nLIGHT ENABLED, SMALL MOTION 360 DEGREE COVERAGE, DAYLIGHT SENSING, INTEGRAL ADJUSTABLE OCCUPANCY SINGLE TIME DELAY/PHOTOCELL SET-POINTS, INTEGRAL LED STATUS INDICATOR, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ HIGH HUMIDITY/DAMP LOCATION RATED.		1
S	ACUITY nLIGHT #NPODM-WH-LT OR APPROVED EQUAL	CAT-5e	SURFACE, RECESSED (OR AS NOTED ON DRAWINGS)	AT 42" A.F.F. (OR AS NOTED ON DRAWINGS)	ON/OFF WALL MOUNTED nLIGHT-ENABLED WALL SWITCH WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, WHITE WALL PLATE, INTEGRAL LED STATUS INDICATOR, INTEGRAL SOFT-CLICK PUSH-BUTTON SINGLE CHANNEL CONTROL, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ DAMP LOCATION RATED.		1
SDM	ACUITY nLIGHT #NPODM-DX-WH-LT OR APPROVED EQUAL	CAT-5e	SURFACE, RECESSED (OR AS NOTED ON DRAWINGS)	AT 42" A.F.F. (OR AS NOTED ON DRAWINGS)	ON/OFF WALL MOUNTED nLIGHT-ENABLED DIMMABLE WALL SWITCH WITH RUGGED HIGH-IMPACT INJECTION-MOLDED WHITE PLASTIC HOUSING, WHITE WALL PLATE, INTEGRAL LED STATUS INDICATOR, INTEGRAL SOFT-CLICK PUSH-BUTTON SINGLE CHANNEL CONTROL, INTEGRAL REMOTELY UPGRADEABLE FIRMWARE, LOW TEMP/ DRY LOCATION RATED.		1
PLF	ACUITY nLIGHT #NPP20PL-LT OR APPROVED EQUAL	120-277V/ 15VDC/ CAT-5e	SURFACE	CEILING/ STRUCTURE	PLUG LOAD CONTROL RELAY POWER PACK WITH RUGGED HIGH-IMPACTED INJECTION-MOLDED BLUE HOUSING, nLIGHT ENABLED, INTEGRAL 120V SIDE SLOT POWER INPUT/OUTPUT, INTEGRAL 15VDC/40mA PER RJ-45 PORT OUTPUT, INTEGRAL EXTERNAL FAULT PROTECTION, INTEGRAL LATCHING RELAY SWITCHES LINE VOLTAGE LOAD, INTEGRAL AUTO ON MODE, INTEGRAL PROGRAMMING BUTTON, INTEGRAL REMOTELY UPGRADABLE FIRMWARE, LOW TEMP/ PLENUM LOCATION RATED.		1
PP	ACUITY nLIGHT #NPP16-DS-EFP-LT OR APPROVED EQUAL	120-277V/ CAT-5e/ 0-10V	SURFACE	CEILING/ STRUCTURE	nLIGHT-ENABLED RELAY POWER PACK WITH RUGGED HIGH-IMPACTED INJECTION-MOLDED WHITE HOUSING, INTEGRAL 0-10V SIDE SLOT DIMMING OUTPUT, INTEGRAL EXTERNAL FAULT PROTECTION, INTEGRAL SELF-CONTAINED RELAY SWITCHES LINE VOLTAGE LOAD, INTEGRAL AUTO ON MODE, INTEGRAL PROGRAMMING BUTTON, INTEGRAL REMOTELY UPGRADABLE FIRMWARE, LOW TEMP/ PLENUM LOCATION RATED.		1

NOTE:
1) COMPLY WITH SELECTED OCCUPANCY/MOTION SENSOR, WALL SWITCH, AND POWER RELAY PACK MANUFACTURER'S INSTALLATION AND FINAL PLACEMENT REQUIREMENTS PRIOR TO ROUGH-IN. APPLICABLE FOR NON-PROCESS AREAS ONLY. REFER TO SPECIFICATIONS FOR LUMINAIRE CONTROL COMPONENTS REQUIRED FOR PROCESS AREAS.

LUMINAIRE CONTROLS SEQUENCE SCHEDULE

SPACE/ AREA TYPE	SEQUENCE OF OPERATIONS (NO DAYLIGHT HARVESTING)	LIGHTING RELAY PANEL	DAYLIGHT HARVESTING - ILLUMINANCE REQUIREMENTS (FC SETPOINTS)
ELECTRICAL ROOM, PUMP ROOM	MANUAL ON/ MANUAL OFF BY LOCAL BI-LEVEL CONTROL DEVICES. OCCUPANT ENTERS THE SPACE AND MANUALLY TURN ON TO 50% OF FULL LUMINAIRES OR 100% OF FULL LUMINAIRES IN THE SPACE.	N/A	N/A
OPERATIONS ROOM	OS - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 50% OF FULL LUMINAIRE OUTPUT AND DEDICATED PLUG LOAD FULL CONTROL 'PLF' RECEPTACLES WILL TURN ON. OCCUPANT CAN MANUALLY TURN ON THE LUMINIARES TO FULL LUMINAIRE OUTPUT BY LOCAL CONTROL DEVICE. UPON SENSING VACANCY AFTER 20 MINUTES, THE LUMINAIRES AND 'PLF' RECEPTACLES AUTOMATICALLY TURN OFF.	N/A	N/A
EXTERIOR WALL PACKS	INTEGRAL - OCCUPANT ENTERS THE SPACE AND THE LUMINAIRES AUTOMATICALLY TURN ON TO 100% OF FULL LUMINAIRE OUTPUT. UPON SENSING VACANCY AFTER 15 MINUTES, THE LUMINAIRES AUTOMATICALLY TURN DOWN TO 30% OF FULL LUMINAIRE OUTPUT.	LIGHTING CIRCUITS TO BE ROUTED THROUGH THE DIGITAL LIGHTING CONTROL RELAY PANEL FOR DIGITAL PHOTOCELL, TIME CLOCK, OR SCHEDULED ON/OFF CONTROL. PROVIDE LOCAL LOW-VOLTAGE MANUAL OVERRIDE SWICH(S) FOR 2 HOUR MAXIMUM OVERRIDE.	N/A

NOTES:

1) REFER TO LIGHTING PLANS AND SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

2) FOR AREAS INDICATED ON THE LIGHTING PLANS TO HAVE DAYLIGHT HARVESTING REQUIREMENTS, THE SEQUENCE OF OPERATIONS FOR THE OCCUPANCY/VANCY SENSOR SHALL BE AS INDICATED FOR THE SPACE/AREA TYPE IN ADDITION TO THE AUTOMATIC DIMMING CONTROL REQUIREMENTS. PROGRAMMING OF THE DAYLIGHT HARVESTING CONTROLS SHALL UTILIZE THE ILLUMINANCE SETPOINTS (FC) INDICATED ON THIS SCHEDULE.

GENERAL NOTE:

A. LUMINAIRE SCHEDULE AND LUMINAIRE CONTROLS SCHEDULE PRESENTS INFORMATION ON THE SALIENT CHARACTERISTIC OF THE LUMINAIRES AND LUMINAIRE CONTROLS. BRAND NAMES LISTED ARE CONSIDERED BASIS OF DESIGN. "OR EQUAL" SUBSTITUTIONS ARE ACCEPTABLE MEETING THE REQUIREMENTS OF 01 33 23 AND THE PRODUCT REQUIREMENTS LISTED IN THE ASSOCIATED TECHNICAL SPECIFICATIONS. ALL "OR EQUAL" SUBSTITUTED LUMINAIRES MUST COMPLY WITH THE PHOTOMETRIC AND POWER DENSITY STATISITICS (ALL INTERIOR AND EXTERIOR SPACES/ AREAS) INDICATED ON THE ELECTRICAL LIGHTING AND RECEPTACLE PLAN(S).

REGISTERED PROFESSIONAL ENGINEERS — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date signed 1/07/2026 — ARIZONA, U.S.A. — ELECTRICAL

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. MARTINEZ DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E631	TITLE OF SHEET RSPH ELECTRICAL LUMINAIRE SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 100 OF 113

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BRANCH PANEL: RSPH-LP-01

LOCATION: ELECTRICAL AREA 103 | **VOLTS:** 208Y/120 | **A.I.C. RATING:** 10KA
SUPPLY FROM: | **wttm** 3 | **MAINS TYPE:** MCB
MOUNTING: SURFACE | **WIRES:** 4 | **MAINS RATING:** 150 A
ENCLOSURE: NEMA 1 | **INTEGRAL SPD:** | **SERVICE ENTRANCE LABEL:** No

WIRE & CONDUIT	CKT	CIRCUIT DESCRIPTION	CLASS	TRIP	P	A	B	C	P	TRIP	CLASS	CIRCUIT DESCRIPTION	CKT	WIRE & CONDUIT
(2) #10, #10 GRD, 3/4"C	1	LGTS RM 101, 102 & 103	Lighting	20 A	1	687.0			1	20 A	REC	REC PUMP ROOM 101	2	(2) #12, #12 GRD, 3/4"C
(2) #12, #12 GRD, 3/4"C	3	LIGHTING CONTROL PANEL RSPH-LCP-01	Power	20 A	1		500.0		1	20 A	REC	REC ELECTRICAL AREA 103	4	(2) #12, #12 GRD, 3/4"C
(2) #10, #10 GRD, 3/4"C	5	LGTS, BLDG EXTERIOR	Lighting	20 A	1			44.0	1	20 A	REC	REC EXTERIOR	6	(2) #12, #12 GRD, 3/4"C
(2) #12, #12 GRD, 3/4"C	7	REC, OPERATIONS ROOM 102	REC	20 A	1	540.0			1	20 A	--	SPARE	8	
(2) #12, #12 GRD, 3/4"C	9	TCP 51741 TEMP CNTRL PANEL 1	Power	20 A	1		1256.0		1	20 A	Power	AFF CONTROL PANEL LCP-52101	10	(2) #12, #12 GRD, 3/4"C
(2) #12, #12 GRD, 3/4"C	11	TCP 51742 TEMP CNTRL PANEL 2	Power	20 A	1			728.0	1	20 A	Power	HYPO XFER PUMP PMP-53421	12	(2) #12, #12 GRD, 3/4"C
(2) #10, #10 GRD, 3/4"C	13	RW PRSR DISSIPATION VALVE LCP-53117	Power	30 A	1	864.0			1	20 A	--	SPARE	14	
	15	SPARE	--	20 A	1		0.0		1	20 A	Power	CL DOSING PUMP PMP-53321	16	(2) #12, #12 GRD, 3/4"C
(2) #12, #12 GRD, 3/4"C	17	HEAT TRACE, FW TANK TK-53211	Power	20 A	1			500.0	1	20 A	Power	CL DOSING PUMP PMP-53322	18	(2) #12, #12 GRD, 3/4"C
(3) #8, #10 GRD, 1"C	19	FW BOOSTER PUMPS LCP-53111	Power	20 A	2	1995.0			1	20 A	Power	FW RECIRC PUMP PMP LCP-53221	20	(2) #12, #12 GRD, 3/4"C
	21						1995.0		1	20 A	--	SPARE	22	(2) #12, #12 GRD, 3/4"C
(2) #10, #10 GRD, 3/4"C	23	PLC CONTROL PANEL	Power	30 A	1			2880.0	1	20 A	--	SPARE	24	(2) #10, #10 GRD, 3/4"C
	25	SPARE	--	20 A	1	0.0	0.0		1	20 A	--	SPARE	26	
	27	SPARE	--	20 A	1		0.0	0.0	1	20 A	--	SPARE	28	
	29	SPARE	--	20 A	1			0.0	1	20 A	--	SPARE	30	
	31	SPARE	--	20 A	1	0.0	1279.0		2	20 A	Power	HVAC	32	(2) #12, #12 GRD, 3/4"C
	33	SPARE	--	20 A	1		0.0	1279.0					34	
	35	SPARE	--	20 A	1			0.0	1	20 A	--	SPARE	36	
	37	SPARE	--	20 A	1	0.0	0.0		1	20 A	--	SPARE	38	
	39	SPARE	--	20 A	1		0.0	0.0	1	20 A	--	SPARE	40	
	41	SPARE	--	20 A	1			0.0	1	20 A	--	SPARE	42	
				TOTAL LOAD:		7141 VA	6938 VA	5256 VA						
				TOTAL AMPS:		62 A	60 A	44 A						

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND
REC	2880 VA	100.00%	2880 VA
Power	15724 VA	100.00%	15724 VA
Lighting	731 VA	125.00%	914 VA

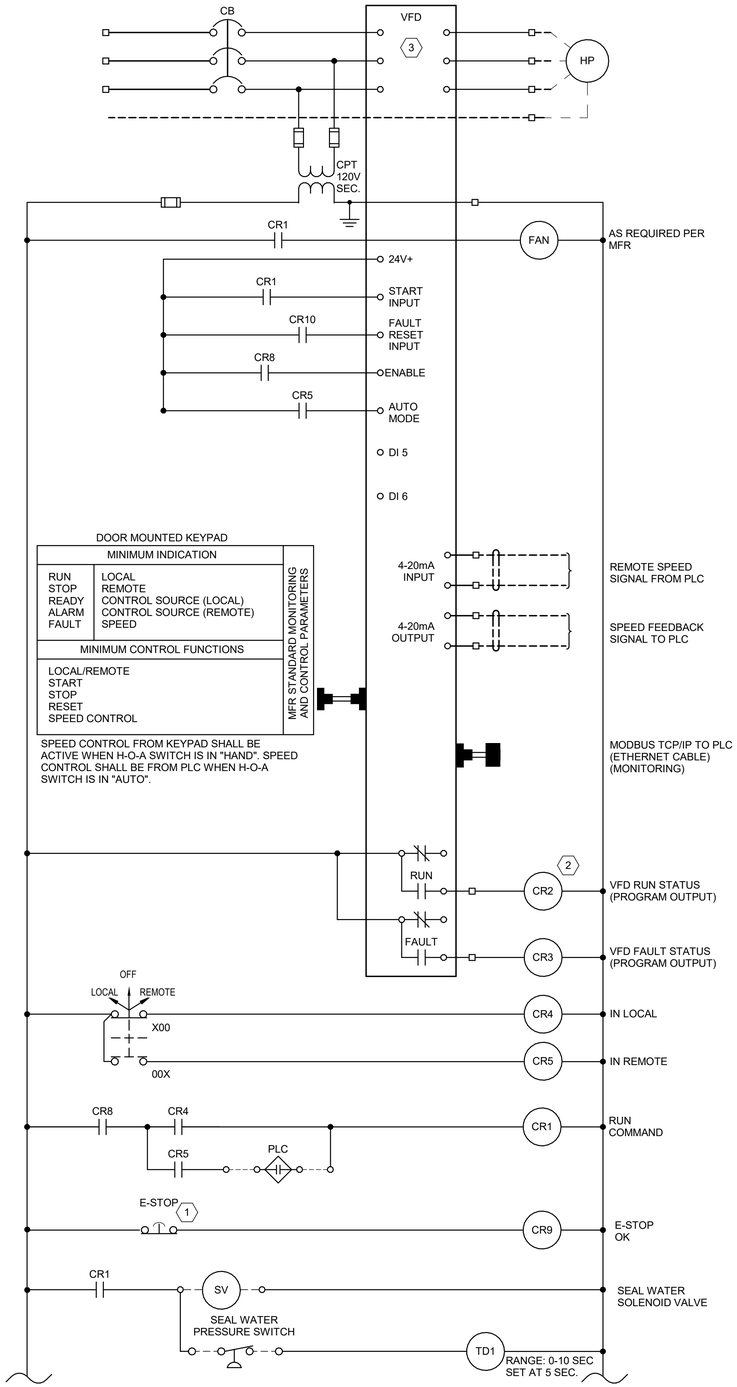
PANEL TOTALS	
TOTAL CONNECTED LOAD:	**19335 VA**
TOTAL ESTIMATED DEMAND:	**19518 VA**
TOTAL CONNECTED:	**54 A**
TOTAL ESTIMATED DEMAND	**54 A**
SPARE AMPERAGE:	
TOTAL ESTIMATED DEMAND + SPARE:	**54 A**

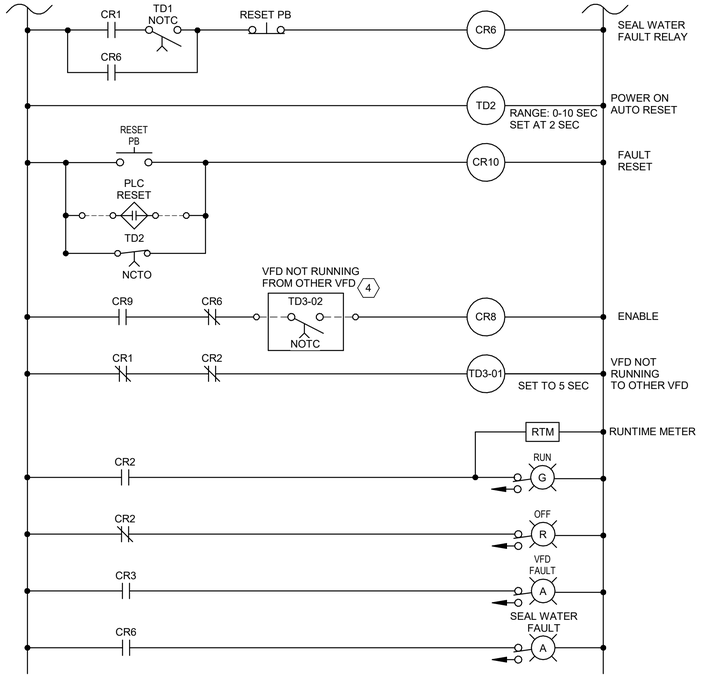
NOTES:
CKT 17 - USE 30 mA EPD TYPE BREAKER

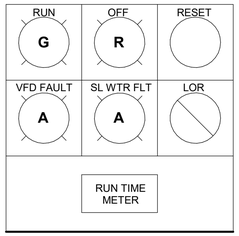
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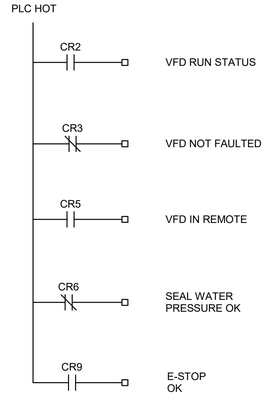
REGISTERED PROFESSIONAL ENGINEER (DEPARTMENT)
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 10/12/2026
ARIZONA, U.S.A.

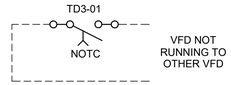
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: T. MOORE DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: AUG 2026	SUB SHEET NO. 052 E632	TITLE OF SHEET RSPH ELECTRICAL PANELBOARD SCHEDULES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 101 OF 113











KEYNOTES (X)

1. RED, MUSHROOM-HEAD, MAINTAINED CONTACT PUSHBUTTON ON FRONT OF VFD ENCLOSURE.
2. SET "VFD RUNNING" OUTPUT TO "ON" SPEED FOR ANY SPEED OTHER THAN ZERO SPEED.
3. REFER TO SPECIFICATIONS 26 35 26 AND 26 29 23 FOR SPECIFIC POWER QUALITY REQUIREMENTS.
4. TIMER TD3-02 IS CONTAINED WITHIN THE PAIRED VFD ENCLOSURE.

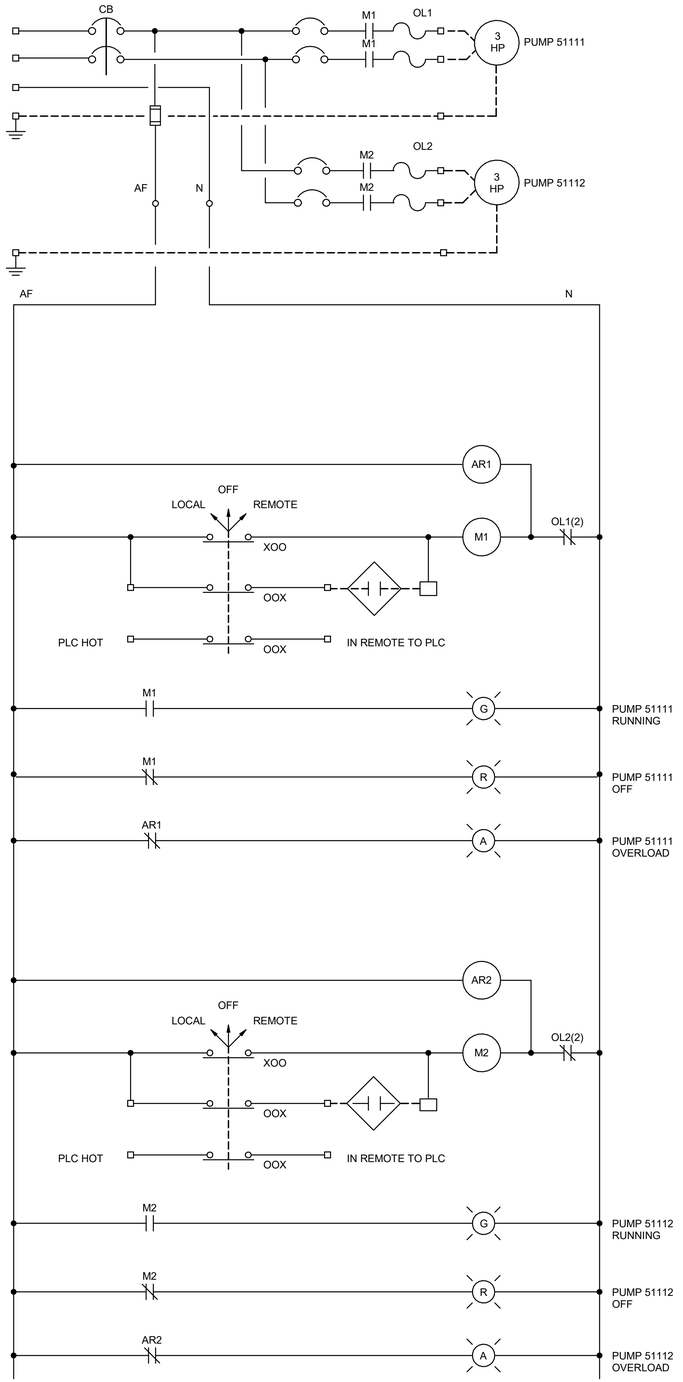
1 / 052E641 — 052-01 RAW WATER PUMPS

PMP-52211, PMP-52212

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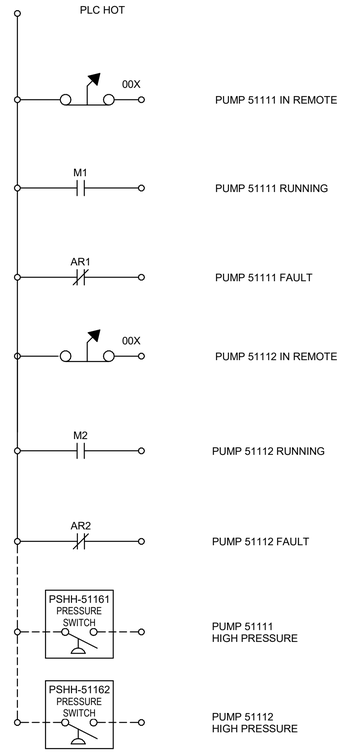
REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 2/18/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER DRAWN: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E641	TITLE OF SHEET RSPH MOTOR CONTROL DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 102 OF 113



1 / 052E642 — 052-02 FINISHED WATER BOOSTER PUMPS

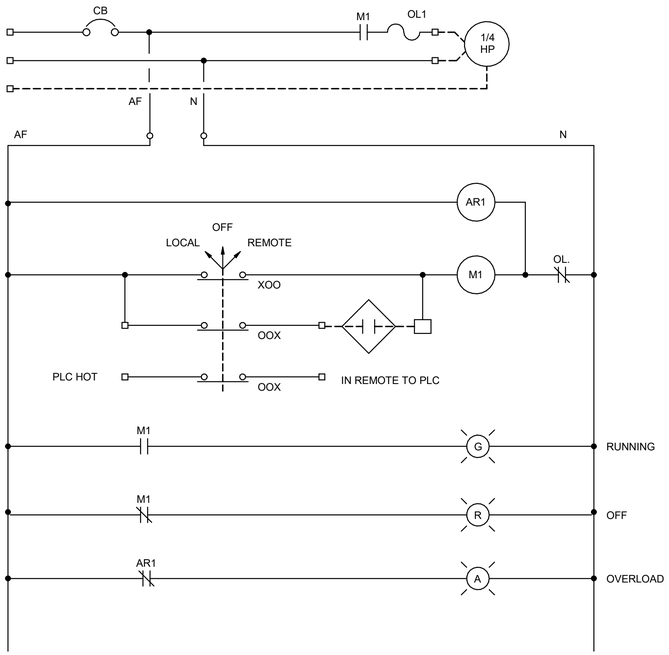
PMP-51111, PMP-51112

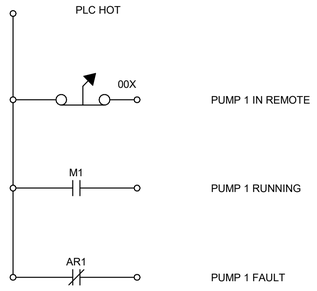


REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 58037 — TRAVIS T. MOORE — Date Signed 4/27/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER DRAFTED: S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E642	TITLE OF SHEET RSPH MOTOR CONTROL DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 103 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_E.rvt
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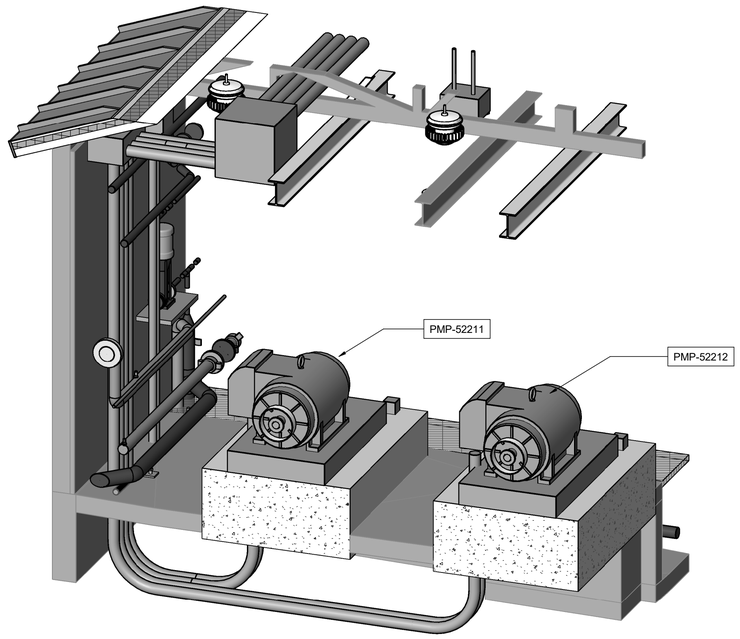
1 / 052E643 — 052-03 FINISHED WATER RECIRC PUMP

PMP-53221

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_E.rvt
4/27/2026 5:28:36 PM

REGISTERED PROFESSIONAL ENGINEER — ELECTRICAL — CERTIFICATE NO. 58037 TRAVIS T. MOORE — Date Signed 4/27/2026 — ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER CADD S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E643	TITLE OF SHEET RSPH MOTOR CONTROL DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 104 OF 113

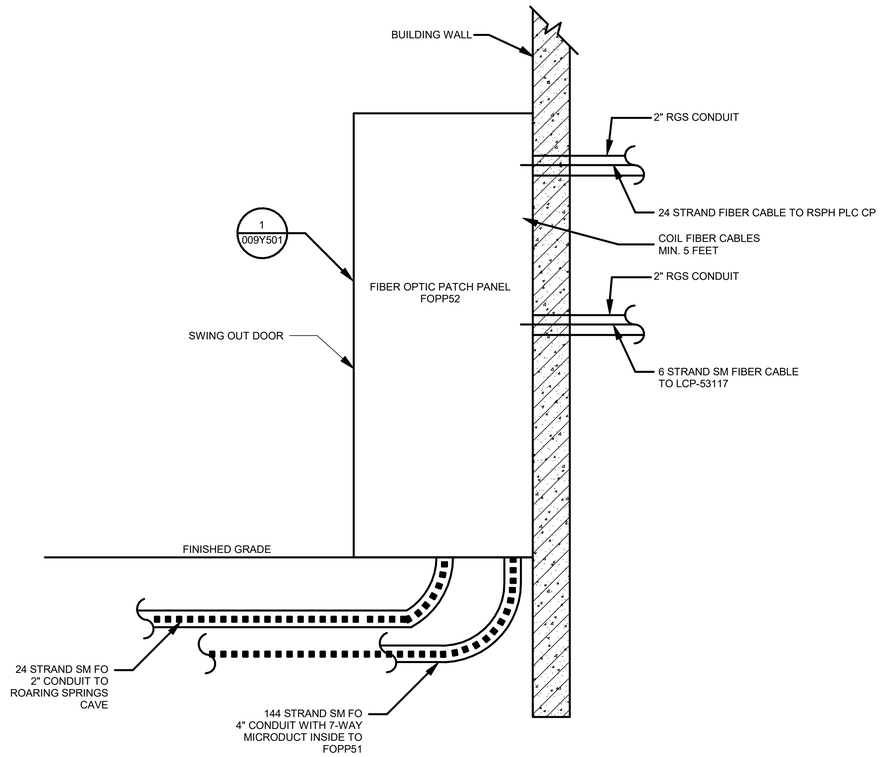


1 (052E901) ISO CONDUITS FOR PMP-52211 AND PMP-52212
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL)
CERTIFICATE NO.
58037
TRAVIS T. MOORE
Date Signed 2/18/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. HUCKENPAHLER CADD S. BILD TECH. REVIEW: P. DAVILA DATE: MAY 2026	SUB SHEET NO. 052 E901	TITLE OF SHEET RSPH ISO CONDUITS FOR PMP-52211 AND PMP-52212 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 105 OF 113

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_E.rvt
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1 / 052Y501 RSPH FIBER BUILDING ENTRY
NOT TO SCALE

Autodesk Docs://10414450_NPS_GRCA_N_Rim_Utility_2026/10350545_052_RSPH_E.rvt
4/27/2026 5:28:50 PM

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 5/1/26
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 052 Y501	TITLE OF SHEET RSPH INSTRUMENTATION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 106 OF 113

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\052Y601.dwg

HMI

RSPH PLC

MCC/CP

PROCESS

TURB AI 52381 | REMOTE HSR 52331 | OPEN CMD XVO 52331 | CLOSE CMD XVC 52331 | OPEN ZSO 52331 | CLOSED ZSC 52331 | FAULT UA 52331 | PRESS PI 52261 | FILTER 1 BACKWASH ACTIVE NI 52111 | FILTER 2 BACKWASH ACTIVE NI 52112 | FAULT UA 52101

FLOW FI 52371 | FLOW TOTAL FQI 52371 | NO FLOW FAL 52371 | PUMP VIB VI 52211A | PUMP BEARING TEMP TI 52211A

ESTOP ES 52211 | SEAL FAIL PAL 52211 | RUN CMD XC 52211 | RUN STATUS NI 52211 | REMOTE HSR 52211 | FAULT UA 52211 | SPEED CMD SC 52211 | SPEED SI 52211 | RESET XR 52211 | MULTI DATA UU 52211 | MOTOR WINDING BEARING TEMP TI 52211B | MOTOR VIB VI 52211D | HIGH HIGH PRESS PAHH 52263

RSPH PLC: AI 52381 | HSR 52331 | XVO 52331 | XVC 52331 | ZSO 52331 | ZSC 52331 | UA 52331 | PI 52261 | NI 52111 | NI 52112 | UA 52101 | FI 52371 | FQI 52371 | FSL 52371 | VI 52211A | TI 52211A | ES 52211 | PSL 52211 | XC 52211 | NI 52211 | HSR 52211 | UA 52211 | SC 52211 | SI 52211 | XR 52211 | UU 52211 | TI 52211B | VI 52211D | PSHH 52263

MCC/CP: I/I 1 — X4: PIT-52261, PIT-52262, PIT-52265, PIT-52266 — X2: FIT-52371, FIT-52372 — X6: VI-52211(2)A, VI-52211(2)B, VI-52211(2)C — X2: TI-52211(2)A — 480V VFD 2 — G 52211 | PB 52211 | RESET | HS 52211 | LOR | I — RTM — X2: PMP-52211, PMP-52212 — X4: G-RUN, R-OFF, A-FAULT, A-SEAL LEAK — X2: RESET, E-STOP — X16: TI-52211(2)B, TI-52211(2)C, TI-52211(2)D, TI-52211(2)E, TI-52211(2)G, TI-52211(2)H, TI-52211(2)J — X12: VI-52211(2)D, VI-52211(2)E, VI-52211(2)F, VI-52211(2)G, VI-52211(2)H, VI-52211(2)J — X2: PSHH-52263, PSHH-52264

PROCESS: AUTO FILTER VENDOR CONTROL PANEL LCP-52101 — 120V — PIT 52262 — PIT 52261 — DPIT 52112 — AUTO FILTER AFV-52112 — V-52132 — V-52134 — RW4 - - FROM RS CAVE — 4"-RW4 — OUTSIDE | RSPH — 1/2"-RW — 480V M FV-52331 — 4"-PD1 — 1 1/2"X2" — M — DPIT 52111 — AUTO FILTER AFV-52111 — 3"X1 1/2" — V-52131 — V-52133 — ARV-52331 — ARV-52332 — 3"-PD1 — 1 1/2"X2" — M — 3"X1 1/2" — AUTO FILTER SKID — 4"X3" — 3"-PD1

FIT 52371 — 120V — M — RAW WATER PUMP SUCTION FLOW METER — 4"X5" — 4"-RW4

VE 52211A-C | TE 52211A | PSL 52211 — V-52231 — M — RAW WATER PUMP 1 PMP-52211 — 2 1/2"X4" — PSHH 52263 — 1/2" — 1/2"-RW5 — CKV-52331 — V-52233 — 4"-RW5

TE 52211B-J | VE 52211D-J

TE 52212B-J | VE 52212D-J — PSHH 52264 — VE 52212A-C | TE 52212A | PSL 52212 — M — 2 1/2"X4" — 1/2"-RW5 — 1/2" — V-52232 — RAW WATER PUMP 2 PMP-52212 — CKV-52332 — V-52234 — 4"-RW5

PIT 52265 — TYP 8 — PIT 52266 — 1/2"-RW5 — 1/2" — CKV-52333 — V-52031 — 4"-RW5 — RSPH | OUTSIDE — V-52032 — RW5 - 062Y601 TO NORTH RIM WTP VIA BOREHOLE (SEE VOL 06)

PI 52366 — 1/2" — 3"X4" — 1/2" — V-52033 — MANUAL CONTROL VALVE EDS-52312

PI 52364 | PI 52365 — ORIFICE PLATE ASSEMBLY — N.O. V-52331 — 3"-PD1

PI 52367 — FIT 52372 — 120V — 3"-PD2 — PD2 A 052Y602 TO RSPH PIPE TRENCH

FE 52372 — 3"X2" — 2"-PD1 — M — RAW WATER BYPASS FLOW METER — 2"X3" — 3"-PD1 — PD1 B 052Y602 TO RSPH PIPE TRENCH

FSL 52371 — 1/2"-PD1

AIT 52381 — 120V — TURB — AE 52381

PI 52361 — 50 PSI — PI 52362 — PRV-52331 — 4 PSI — 3 — PI 52363 — FI 52371 — 4 | 4

INSTRUMENTATION PANEL 7

GENERAL NOTES:

A. * - EQUIPMENT PROVIDED BY TURBIDITY METER VENDOR.

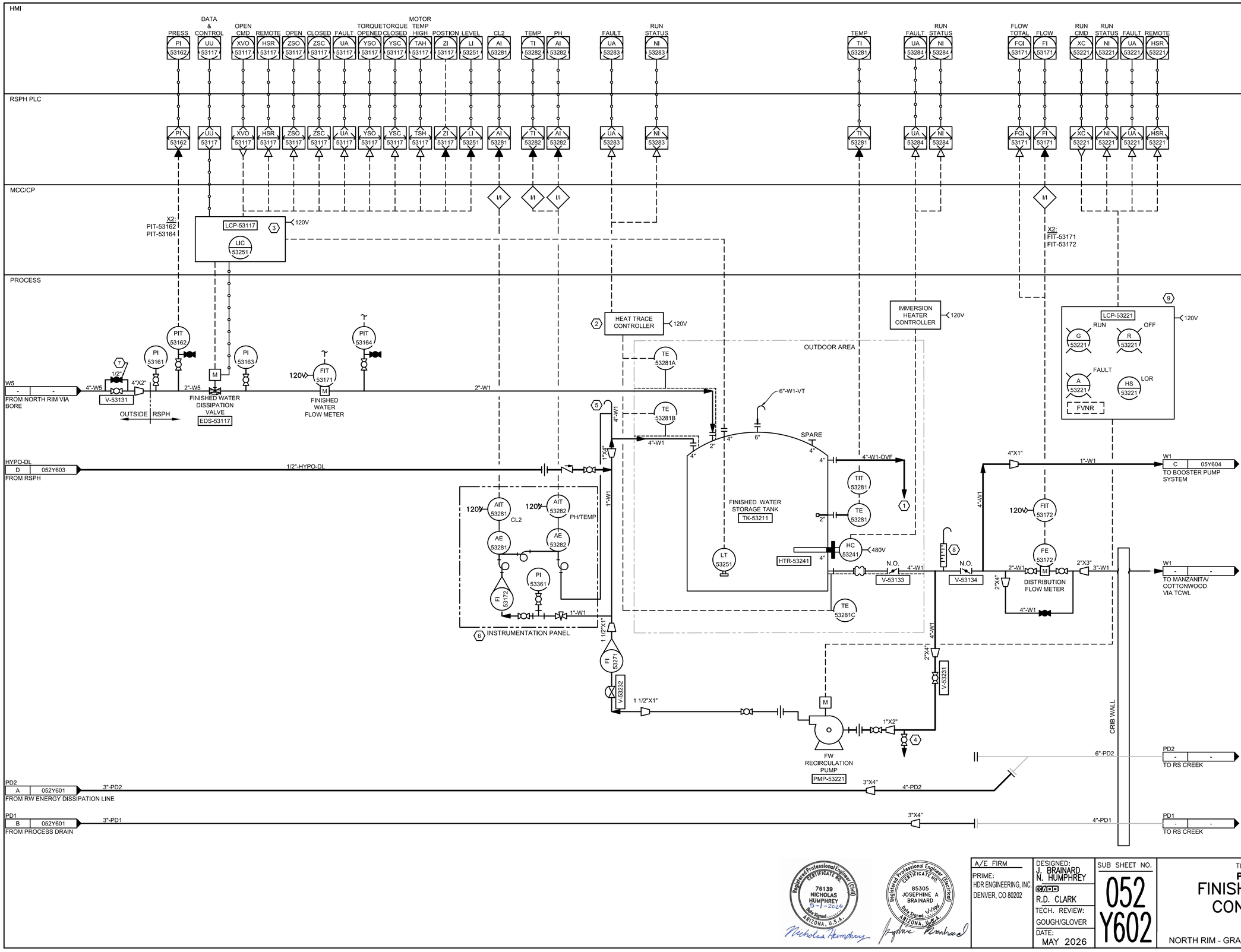
KEYNOTES: (X)

1. SIGNAL ISOLATORS TO BE INSTALLED INSIDE PLC CONTROL PANEL AND POWERED WITH 24VDC TYPICAL FOR ALL.
2. SEE CONTROL DRAWING 1/052E641.
3. OPTIONAL ADJUSTABLE PRESSURE REGULATOR.
4. AIR GAP TO GRAVITY COMMON DRAIN.
5. AIR GAP.
6. TAP SIDE OF PIPE FOR TURBIDIMETER CONNECTION.
7. SEE INSTRUMENTATION PANEL DETAIL ON 052D502.
8. PRESSURE EQUALIZING BYPASS PLUMBED TO ALL HIGH PRESSURE ISOLATION VALVES.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A. — Nicholas Humphrey

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A. — Josephine Brainard

A/E FIRM: PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD N. HUMPHREY; DRAWN: R.D. CLARK; TECH. REVIEW: GOUGH/GLOVER; DATE: MAY 2026	SUB SHEET NO. 052 Y601	TITLE OF SHEET: PUMPHOUSE RAW WATER PUMPING P&ID — NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752; PMIS/PKG NO. 318719; SHEET 107 OF 113	



KEYNOTES: (X)

1. FINISHED WATER STORAGE TANK OVERFLOW TO GROUND.
2. INSULATED PIPING WITH HEAT TRACE AND INDEPENDENT THERMOCOUPLES EA PIPE WIRED TO LOCAL HEAT TRACE CONTROLLER.
3. SEE PANEL LAYOUT DRAWING 052Y612 AND POWER SCHEMATIC 052Y613.
4. FINISHED WATER TANK DRAIN, SEE DETAIL 1/052D501.
5. AIR GAP.
6. SEE INSTRUMENTATION PANEL DETAIL ON 052D502.
7. PRESSURE EQUALIZING BYPASS PLUMBED TO ALL HIGH PRESSURE ISOLATION VALVES.
8. VISUAL TANK LEVEL INDICATOR, SEE 052D309.
9. SEE CONTROL DRAWINGS 1/502E643.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

Registered Professional Engineer (Emeritus) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM: PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD N. HUMPHREY	SUB SHEET NO. 052 Y602	TITLE OF SHEET: PUMPHOUSE FINISHED WATER CONVEYANCE P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752	
	ADD R.D. CLARK			PMIS/PKG NO. 318719	
	TECH. REVIEW: GOUGH/GLOVER			SHEET 108 OF 113	
	DATE: MAY 2026				

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\052Y602.dwg

HMI

OPEN CMD XVO 53331 | CLOSE CMD XVC 53331 | OPEN ZSO 53331 | CLOSED ZSC 53331

TOTAL FLOW FQI 53371 | FLOW FI 53371

RUN CMD XC 53421 | RUN STATUS NI 53421 | REMOTE HSR 53421 | FAULT UA 53421 | SPEED CMD SC 53421 | SPEED SI 53421 | HOSE LEAK MAH 53421

WEIGHT WI 53491 | HIGH LEVEL LAH 53451 | LEVEL LI 53351

RUN CMD XC 53321 | RUN STATUS NI 53321 | REMOTE HSR 53321 | FAULT UA 53321 | SPEED CMD SC 53321 | SPEED SI 53321 | HOSE LEAK MAH 53321

HIGH HIGH PRESS PAHH 53361

RSPH PLC

XVO 53331 | XVC 53331 | ZSO 53331 | ZSC 53331

FQI 53371 | FI 53371

XC 53421 | NI 53421 | HSR 53421 | UA 53421 | SC 53421 | SI 53421 | MSH 53421

WI 53491 | LSH 53451 | LI 53351

XC 53321 | NI 53321 | HSR 53321 | UA 53321 | SC 53321 | SI 53321 | MSH 53321

PSHH 53361

MCC/CP

I/I | I/I | I/I | I/I

X2:
PMP-53321
PMP-53322

X2:
PSHH-53361
PSHH-53362

PROCESS

W1
E 052Y604
FROM BOOSTER PUMP SYSTEM

1"-W1

120V M

FV-53331 | V-53331

FIT 53371 120V M
MAKEUP WATER FLOW METER

1"-W1

1/2"-HYPO-DL

LCP-53421
HS 53421 LOR 2 M 120V
MSH 53421

BATCH METERING PUMP
PMP-53421

LIT 53351

1

2"-OF

LE 53351

LCV-53332

6% - 12.5% STOCK HYPOCHLORITE DRUM OR CARBOY
SCL-53411

120V WIT 53491

DILUTE SODIUM HYPOCHLORITE TANK
TK-53311

1"X2"

2"-HYPO-DL-DR

SECONDARY CONTAINMENT
ENC-53401

LSH 53451

1/2"-HYPO-DL
V-53333

1/2"-HYPO-DL

LCP-53321
HS 53321 LOR 2 M 120V
MSH 53321

PI 53361 | PSHH 53361

35 PSI

METERING PUMP 1
PMP-53321

LCP-53322
HS 53322 LOR 2 M 120V
MSH 53322

PI 53362 | PSHH 53362

35 PSI

METERING PUMP 2
PMP-53322

25 PSI

HYPO-DL
D 052Y602
TO W1 DOSING

CHEMICAL METERING PUMP SKID PROVIDED BY VENDOR

GENERAL NOTES:

A. ALL SODIUM HYPOCHLORITE SERVICE BALL VALVES TO BE VENTED TYPE.

KEYNOTES: (X)

1. TANK VENT INTEGRAL TO TANK LID.
2. HSA SWITCH IS INTEGRAL TO METERING PUMP.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD N. HUMPHREY CADD R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 Y603	TITLE OF SHEET PUMPHOUSE CHLORINE DOSING P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 109 OF 113

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 – NPS GRCA N. RIM UTILITIES\PID DWG\052Y603.dwg

HMI

RSPH PLC

MCC/CP

PROCESS

RUN CMD XC 53111
RUN STATUS NI 53111
FAULT UA 53111
REMOTE HSR 53111
HIGH PRESS PAH 53111
TOTAL FLOW FQI 53173
FLOW FI 53173
PRESS PI 53161
ESEW IN USE FAH 51671

XC 53111
NI 53111
UA 53111
HSR 53111
PSH 53111
FQI 53173
FI 53173
PI 53161
FSH 51671

I/I

X2:
PMP-53111
PMP-53112

208V/1P

LCP-53111

RUN G 53111
OFF R 53111
RUN G 53112
OFF R 53112
FAULT A 53111
HS 53111 LOR
FAULT A 53112
HS 53112 LOR
FVNR
FVNR

1

PSH 53111

PIT 53161

PROCESS | MECHANICAL

≤ 80 PSI

EXPANSION TANK
TK-51641

480V

2"-W1

WH-51641

TMV-1

1"-W1

FSH 51671

120V

ESEW-51641

MECHANICAL | PROCESS

1"-W1

W1
E 052Y603
TO DILUTE HYPO SYSTEM

PI 53166

TK-53111
BLADDER TANK

2"-W1

120V
FIT 53173 M
FW BOOSTER SYSTEM FLOW METER

PI 53165

1"-W1

1"-W1

2"-W1

M
FW BOOSTER PUMP 1
PMP-53111

M
FW BOOSTER PUMP 2
PMP-53112

PSH 53112

W1
C 052Y602
FROM FINISHED WATER TANK

1"-W1

KEYNOTES: (X)

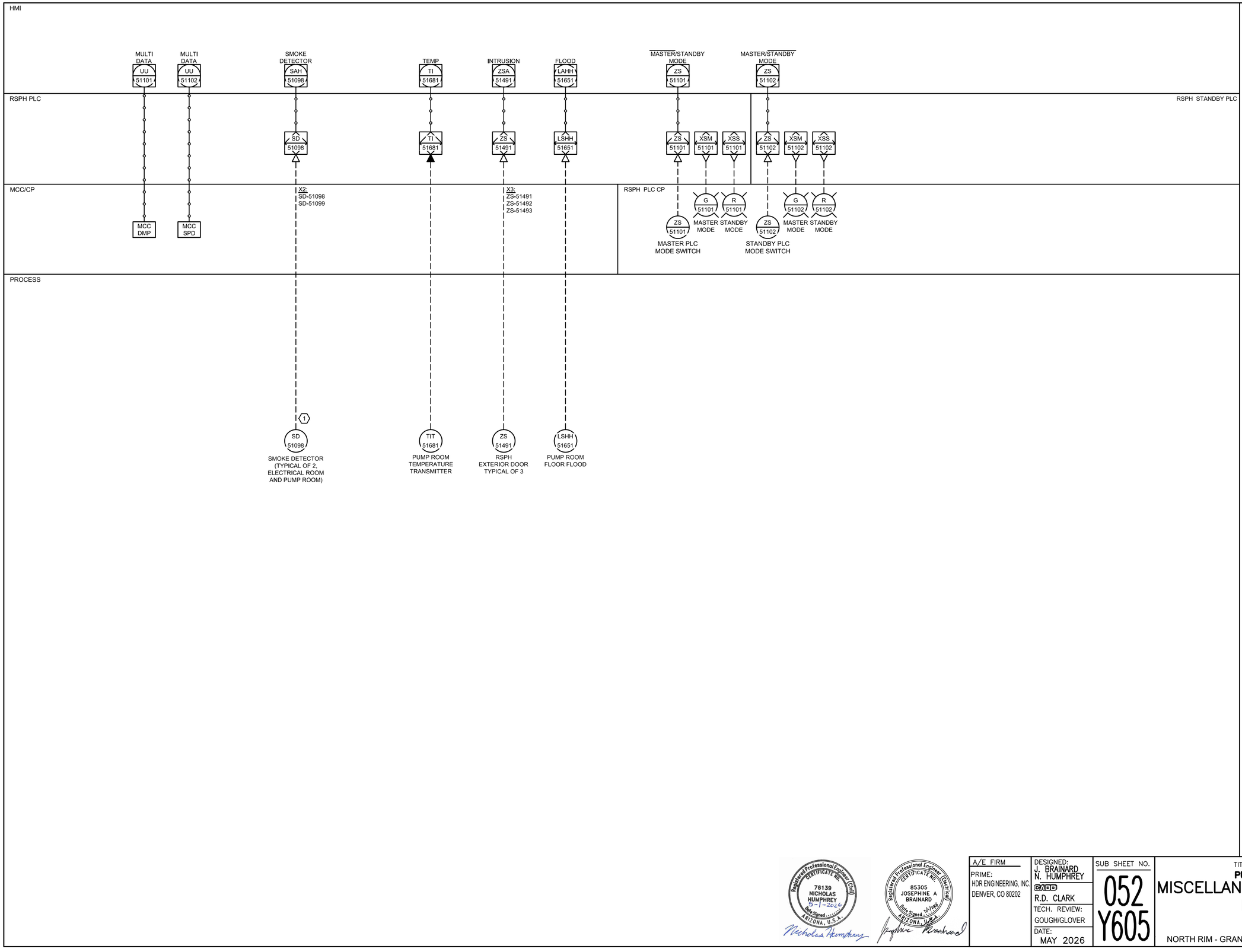
1. SEE CONTROL DRAWING 1/052E642.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD N. HUMPHREY DRAFTED: R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 Y604	TITLE OF SHEET PUMPHOUSE FINISHED WATER BOOSTER PUMPS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 110 OF 113

4/27/2026 Lynch, Nicole C:\Users\nlynch\AppData\Local\Autodesk\AutoCAD Plant 3D\CollaborationCache\10350545 - NPS GRCA N. RIM UTILITIES\PID DWG\052Y604.dwg



KEYNOTES: (X)

1. SMOKE DETECTORS ARE FOR ASSET PROTECTION ONLY. NOT FOR FIRE & LIFE SAFETY. CONTRACTOR TO PROVIDE 120VAC DUCT TYPE SMOKE DETECTOR.

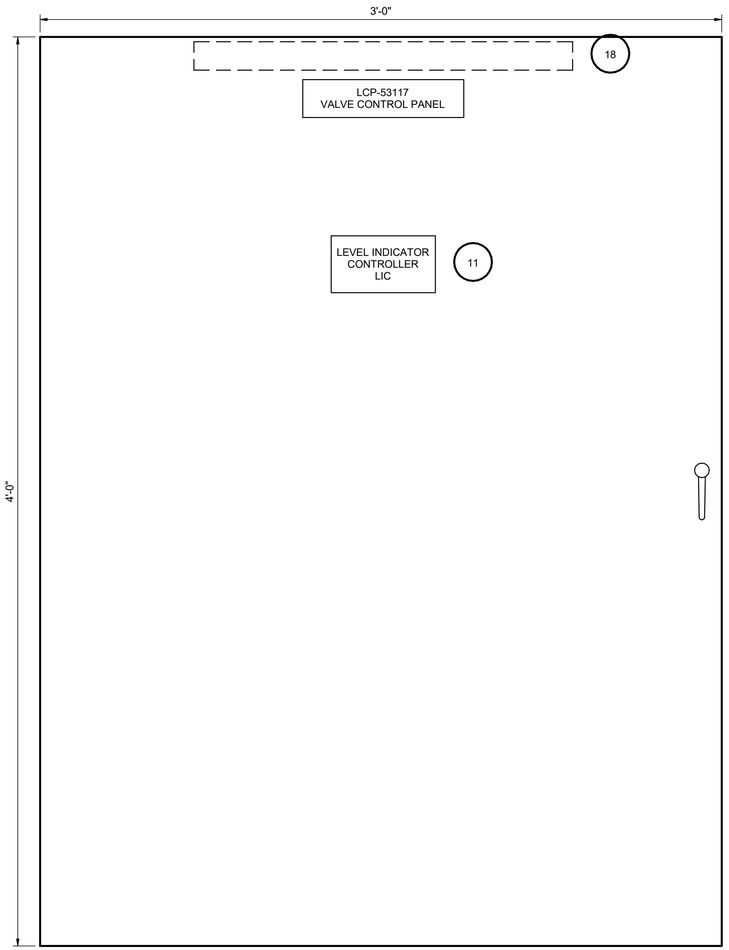
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 5/1/26 ARIZONA, U.S.A.

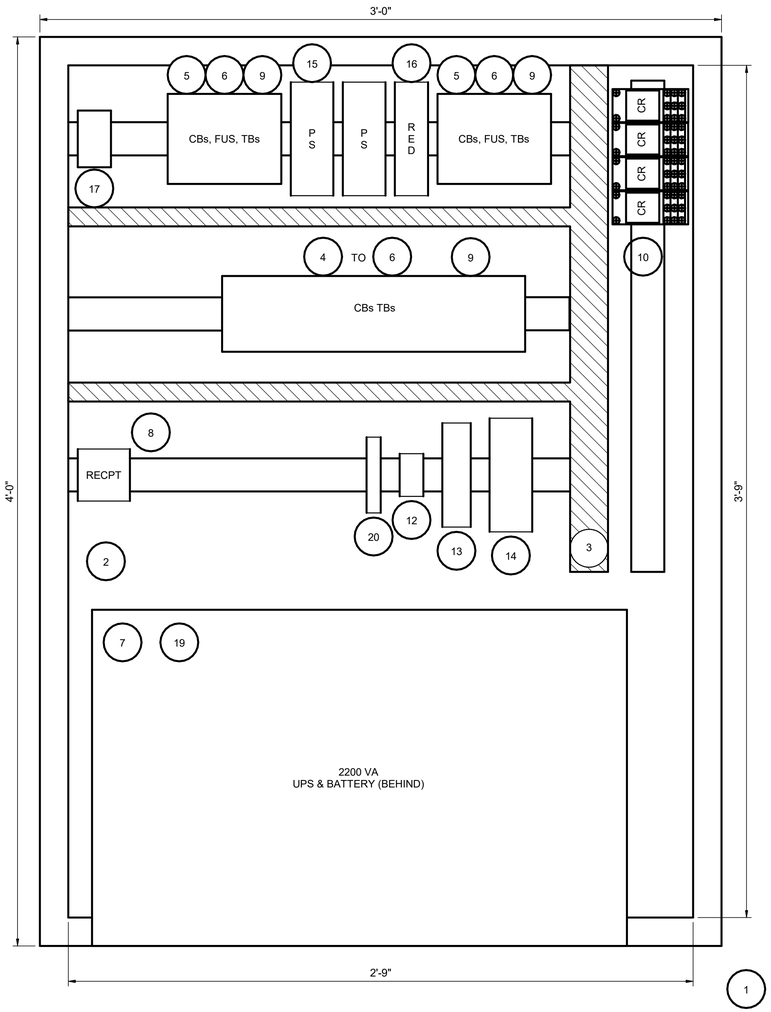
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD N. HUMPHREY CADD R.D. CLARK TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 052 Y605	TITLE OF SHEET PUMPHOUSE MISCELLANEOUS SYSTEMS P&ID NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 111 OF 113	

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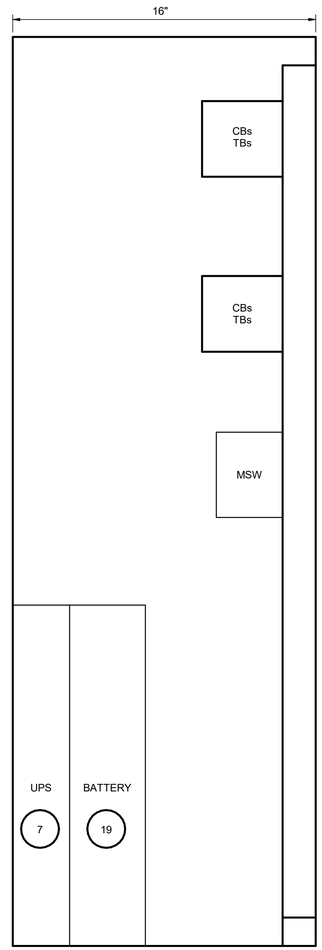
RSPH VALVE CONTROL PANEL BOM		
ITEM #	QTY	DESCRIPTION
1	1	NEMA 12 ENCLOSURE
2	1	ENCLOSURE BACKPANEL
3	AS REQ'D	WIREWAY
4	1	MAIN POWER CIRCUIT BREAKER
5	AS REQ'D	CIRCUIT BREAKERS, SIZE AS REQ'D
6	AS REQ'D	FUSES
7	1	UPS
8	1	UPS RECEPTACLE
9	AS REQ'D	TERMINAL BLOCKS
10	AS REQ'D	CONTROL RELAYS
11	1	LEVEL INDICATOR CONTROLLER
12	1	M-LINK TO SERIAL CONVERTER
13	1	SERIAL TO ETHERNET GATEWAY
14	1	MANAGED ETHERNET SWITCH
15	2	POWER SUPPLY
16	1	REDUNDANCY MODULE
17	1	GFCI RECEPTACLE
18	1	ENCLOSURE LIGHT
19	1	BATTERY PACK
20	1	TIME RELAY



1 052Y612 **FRONT ELEVATION**
NOT TO SCALE



2 052Y612 **INTERIOR ELEVATION**
NOT TO SCALE

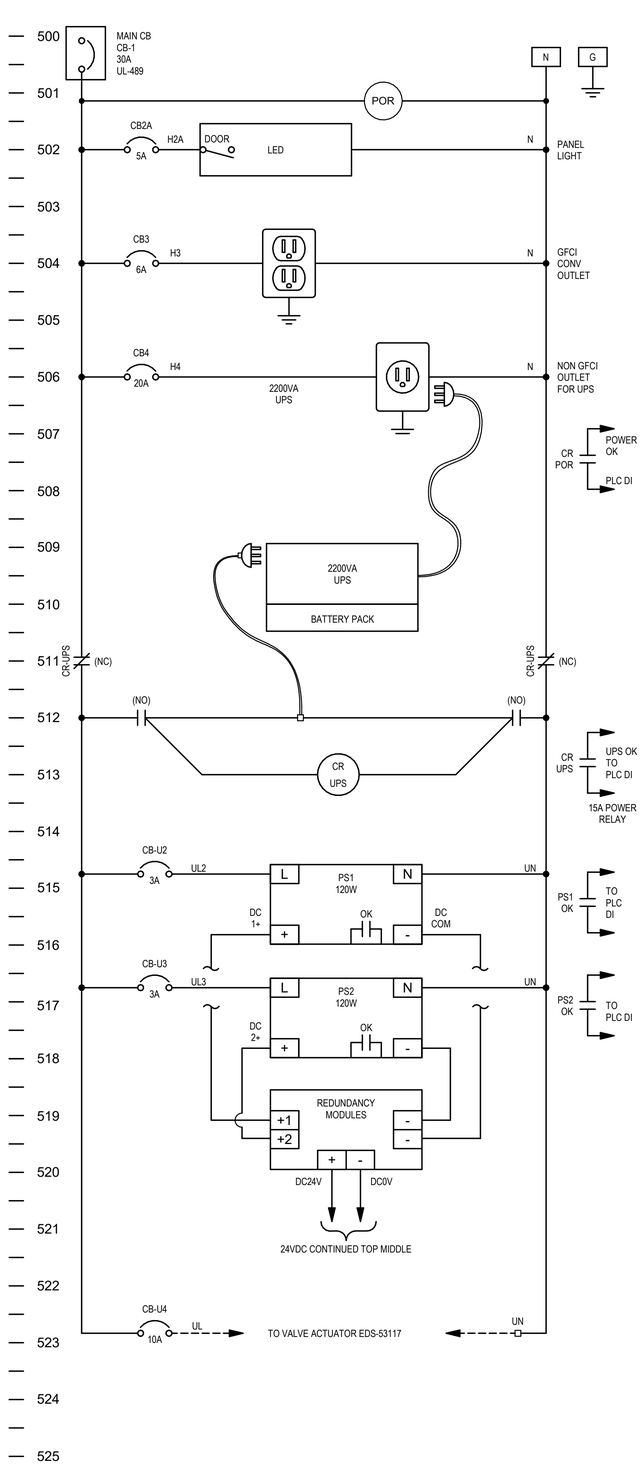


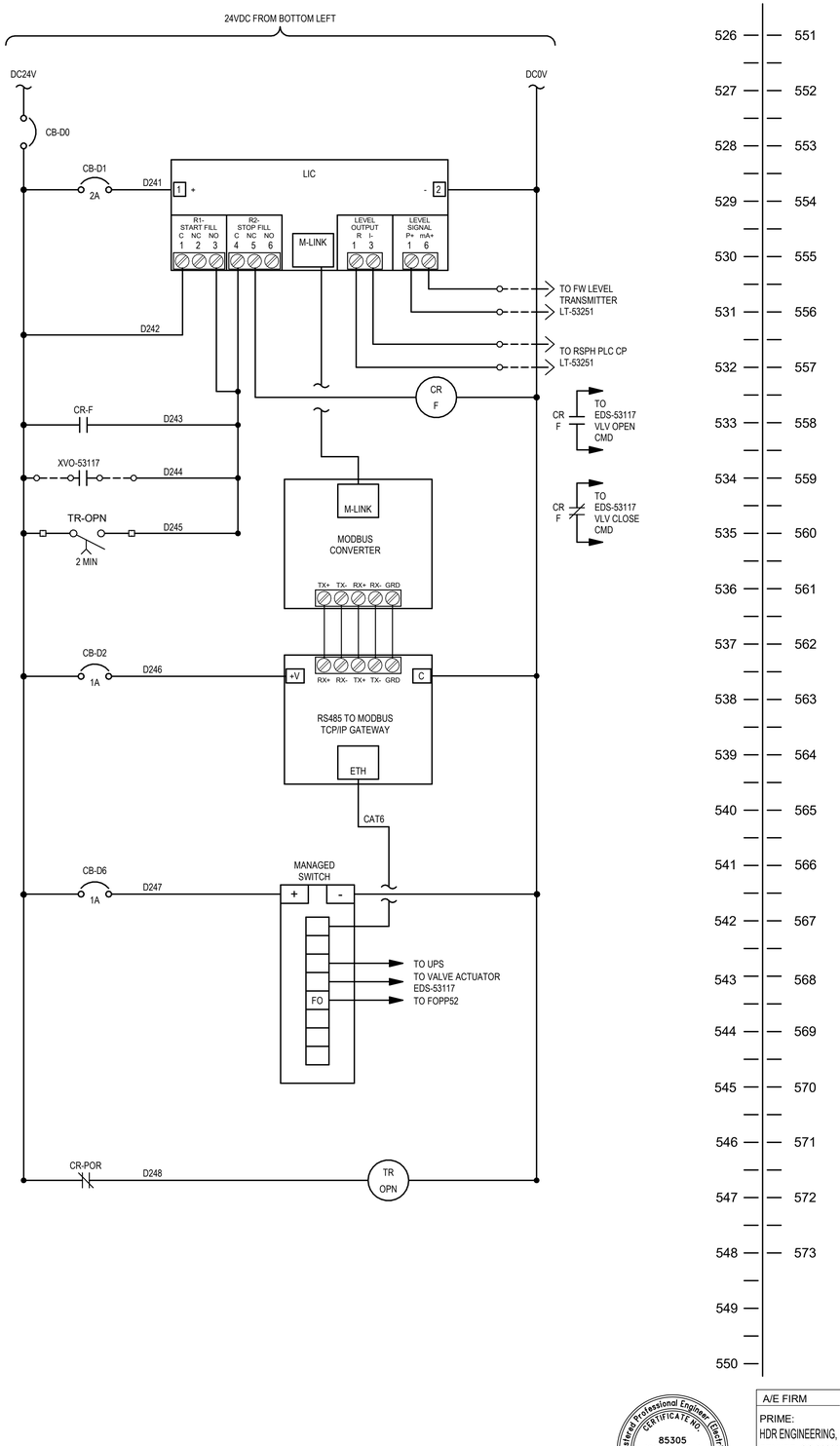
3 052Y612 **RIGHT SIDE INTERIOR ELEVATION**
NOT TO SCALE

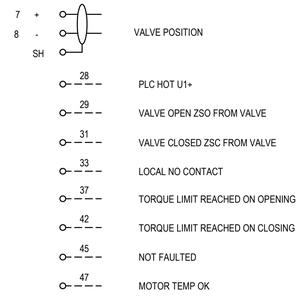
Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A
BRAINARD
Date Signed 04/17/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 052 Y612	TITLE OF SHEET RSPH LCP-53117 VALVE CONTROL PANEL LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 112 OF 113

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GENERAL NOTES

A. SCHEMATIC IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MANUFACTURER'S INSTRUCTIONS.

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Registered Professional Engineer (Electrical)
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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: S. BILD TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 052 Y613	TITLE OF SHEET RSPH LCP-53117 VALVE CONTROL PANEL POWER SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 113 OF 113